

Derek Martin Final ePortfolio

MIME 498 Capstone Design II | Oregon State University | June 2026

Submission Evidence and Compiled Artifact Packet

This compiled PDF includes proof that each required final ePortfolio artifact has been added and is organized for review. The website in this repository also links each artifact from its corresponding portfolio page.

Required item	Included artifact
Resume	derek-martin-resume-google-export.pdf
Cover letter	derek-martin-cover-letter-evensol.pdf
Individual scope/research statement and design proposal	derek-martin-design-proposal.pdf
Individual self-learning assignment	derek-martin-self-learning-assignment.pdf
Final team report	team044-pianobot-final-drawing-package.pdf
Final team report supporting document	team044-pianobot-design-changes.pdf
Team charter	derek-martin-team-charter-clean.pdf
Reflection essay mapping capstone experience to NACE competencies and future goals	derek-martin-reflection-essay-final.pdf
Final team BOM	team044-pianobot-final-bom.xlsx linked in the online portfolio

Artifact 1: Resume

Required item: Resume

Included file: derek-martin-resume-google-export.pdf

Current resume PDF exported from the live Google Docs source.

Derek Martin

Bend, Oregon | (541) 213-1776 | Derekmartin3737@gmail.com

Personal Summary

Mechanical Engineering student (aerospace focus, physics minor) with strength in space systems thinking and hands-on mechanical design. Built Python automation tools and currently developing Python control software (and supporting CAD/integration) for an autonomous piano-player capstone (target delivery June 2026). Designed and analyzed a high-safety-factor aluminum winch stand to lift a 150 lb motor ~50 ft. Seeking remote part-time engineering work through June 2027 and launch/space roles starting 2027.

Education

B.S. in Mechanical Engineering (Focus: Aerospace Engineering, Minor: Physics)

Oregon State University | Corvallis, Oregon

B.S. Expected Graduation: June 2026

Master's in Engineering Expected Graduation: June 2027

GPA: 3.76 & Honor Roll Every Term

Relevant Coursework: Space Systems Engineering, Systems Dynamics & Control, Dynamics, Heat Transfer, Mechanics of Materials, Instrumentation

Professional Experience

Engineering Intern | Evensol Inc. - EDG-L (Landfill Gas Collection) | June 2024 - September 2024

- Conducted equipment inspections, troubleshooting, and preventative maintenance on critical systems.
- Utilized Autodesk Fusion 360 to design and simulate a hand crank project for motor maintenance, gaining skills in 3D modeling and design evaluation.
- Supported feasibility studies and project pricing; contributed to cost estimates and documentation.
- Developed and maintained data-tracking spreadsheets; used Excel for analysis and reporting to support operational decision-making.
- Gained hands-on experience with SCADA systems, transmitter calibration, and gas sampling techniques.
- Collaborated across disciplines on site projects (installations, troubleshooting, and infrastructure improvements) in a safety-focused environment.

Wildland Firefighter | USFS Sisters Ranger District | August 2022 - September 2023

- Implemented fire suppression techniques, including direct and indirect fire line construction, demonstrating strong problem-solving under pressure.
- Operated various firefighting equipment and maintained chainsaw certification for hazardous tree removal and vegetation clearing.
- Collaborated effectively within a high-stakes team environment, utilizing navigation and GPS tools for efficient wildfire suppression.

Technical Skills

Software: Python (automation/control), Microsoft Excel, 3D Modeling (Autodesk Fusion 360), SCADA

Safety Protocols: PPE, working at heights, emergency response

Project Management: Evaluation, scheduling, resource management, data analysis and documentation

Systems: Space systems fundamentals, instrumentation exposure, reliability & safety mindset

Certifications

Emergency Medical Technician (EMT) | Certified through Impact EMS | June 9, 2024

Achievements

- Awarded "Hardest Worker" in Calculus (2021)
- Team Captain for Nordic Skiing
- Medallion awarded by Bend Police Department for integrity and discipline (2021)

Projects

- Autonomous Piano Player (Capstone, 2025–2026): Developing Python control software for an electromechanical system; supporting CAD/integration as needed.
- Winch Lift Stand: Designed and analyzed an aluminum stand supporting a winch to raise a 150 lb motor ~50 ft with a high factor of safety.
- Food Stirrer Prototype: Performed stress analysis and supported iterative design for a 3D-printed liquid-food stirrer with a 4-person team.
- Python Automation: Built a "Jarvis" assistant integrating calendar + smart-home control + weather/maps services; developed a versioned chess bot (~1000 Elo).

Artifact 2: Cover Letter

Required item: Cover letter

Included file: derek-martin-cover-letter-evensol.pdf

Targeted Evensol cover letter.

Derek Martin

Mechanical Engineering—Physics Minor—Aerospace Focus

Derekmartin3737@gmail.com

April 30, 2026

Evensol Hiring Team

Evensol LLC

Sisters, Oregon

Dear Evensol Hiring Team,

I am currently working towards a degree in mechanical engineering from Oregon State University with a physics minor focusing on aerospace engineering. I have previously completed an internship at Evensol, and I wish to return to the company due to its focus on practical engineering, renewables, field reliability, and rigorous project execution. Having interned at Evensol, I have gained insight into the level of technical expertise that is required to work with renewables, and I find great interest in making contributions to biogas and renewable natural gas production.

Throughout my internship, I helped in the execution of engineering tasks such as inspection of the equipment, troubleshooting, preventive maintenance, analysis of project feasibility, cost estimations, and data recording. I have learned to organize data in spreadsheets in order to inform decisions made in the projects in which I worked. This experience has taught me about mechanical engineering projects in their broader operating environment in which the engineering decision-making process plays a significant role.

Moreover, I possess several competencies in engineering, specifically related to the work performed by Evensol. I have modeled and analyzed the designs of various mechanical parts using Autodesk Fusion 360 software, and I designed a winch stand with a high safety factor for lifting a 150 lb motor and a custom hand crank. In addition, I created automation and control programs for engineering projects using Python coding. These experiences have allowed me to develop skills in handling technical problems in mechanical engineering through modeling and documentation.

Apart from the above, I have developed a unique set of professional skills during my previous career experiences as a firefighter and EMT. Being an emergency medical technician, I learned how to communicate, operate according to safety protocols, and perform dependably while being part of a team. These skills can prove useful in performing work related to engineering tasks and projects.

Thank you for considering my application. Please let me know if there will be an opportunity to discuss my qualifications further.

Sincerely,

Derek Martin

Artifact 3: Individual Design Proposal

Required item: Individual scope/research statement and design proposal

Included file: derek-martin-design-proposal.pdf

Design proposal for Musical Bot Encore.

Design Proposal for MIME Capstone Design

February 27, 2026

Team #618

Musical Bot Encore

Musical Bot Encore: The Modular Keyboard-Playing Robot

Derek Martin

School of Mechanical, Industrial, and Manufacturing Engineering

Oregon State University

1. Introduction and Project Scope

Engineering Expo is one of the most prominent outreach events organized by the College of Engineering. For many K-12 attendees, this event represents their first hands-on experience with a robot built by a student. The historic GuitarBot exhibit has served this role admirably in the past; however, its age, wear, and tear on sound quality have diminished its effectiveness. Musical Bot Encore aims to replace this exhibit with a novel electromechanical system that mounts on a household keyboard to play songs autonomously. Our goal is to create an exhibit that is reliable, portable, visually interesting, and able to withstand the rigors of multiple trips to Engineering Expo to captivate K-12 attendees and inspire them to become future College of Engineering students who can appreciate the intersection of mechanical engineering, electrical engineering, and software engineering.

The system takes pre-selected digital music and translates it into mechanical key presses. A software pipeline decodes the music data, an Arduino controller manages the schedule of note presses, driver circuitry controls the required currents, and an array of solenoids presses the keys through a non-invasive frame. The system's use of an actual keyboard to play music instead of a display screen helps to preserve the cause-and-effect connection between the system's actions and its demonstration of mechanical engineering principles that makes the system more memorable to attendees while still being replicable with simple electronics, 3D-printed parts, and open-source design tools.

1.1 Scope Definition

The scope of this proposal is to design a robot to play a keyboard, not a production-quality autonomous music system. The scope of this project includes designing a modular robot to play complete songs without human intervention after initialization, to mount on a keyboard without modification to the keyboard, to assemble quickly to allow for multiple demonstrations at public events such as Engineering Expo, to be portable, visually interesting, and to be built on a student budget. The design should also be extensible to allow future teams to expand the repertoire of songs beyond the initial set of songs chosen for demonstration and perhaps to incorporate additional features such as more keys or sustain pedal support in the future.

1.2 Stakeholders and Customer Requirements

The major stakeholders for this project include Dr. Sarah Oman as the project sponsor, MIME

Capstone instructional staff, Oregon State University's College of Engineering, Engineering Expo staff, and future capstone teams that may maintain or expand the system. The public would include students in grades K-12 and their families. Therefore, the design would need to be technically sound yet accessible to this group. The customer requirements that the team has determined include electronically driven autonomous playing, correct note play, nice sound, aesthetically creative exterior design, easy and interactive design, wide note and dynamic capability, non-invasive securing system, portability, and fast setup. These requirements would be quantitatively defined as shown in Table 1. For this project, a successful song run would be considered one that runs from start to finish without any failure that would negate the functionality of the system. It would not require any human input after the start command.

Table 1. Customer requirements translated into engineering specifications.

Specification	Target	Why It Matters
Autonomous operation	>=95% successful song runs over 20 trials; 0 human inputs after start	Primary reliability metric for Engineering Expo use
Song library	At least 3 playable songs	Keeps the exhibit varied and repeatable
Material reuse	>=50% recycled materials by part count	Supports sustainability goals
Setup and portability	<10 min setup and <100 lb total system mass	Enables transport and rapid event deployment
Mounting method	Non-invasive attachment to different home keyboards	Protects equipment and broadens compatibility
Performance quality	Correct note playback with pleasant sound and visible actuation	Maintains technical credibility and audience appeal

1.3 Background and Motivation

Past research on robotic musicianship has revealed that a computerized music performance system is a viable educational tool, yet also poses a complex engineering problem. The development of robotic percussion and keyboard systems has clearly shown that proper note triggering is achievable, yet making a convincing music performance is more than simply striking the right note [1], [2]. The study on robotic piano music performances revealed that timing, articulation, and dynamic variations significantly affect a music performance, making it sound more human or machine-like [3]-[5]. This means that for Musical Bot Encore, the project does not require concert-level expressiveness to succeed as an outreach exhibit, yet we need to be able to control timing and striking intensity well enough to avoid a machine-like sound.

2. Constraints, Risks, and Assumptions

2.1 Constraints

The key constraints in the project are:

- The system should be ready for Expo within the constraints of the MIME 497/498 timeline.
- The system should be compatible with a student budget of about \$1,000.
- The system should be non-invasively attachable to a home keyboard, requiring no permanent modification to the keyboard.
- At least 50% of parts should be recycled or reused.
- The system should be portable, require less than 10 minutes to set up, and weigh less than 100 lb.

2.2 Risks

The major technical risk here is that there may be insufficient actuation speed or consistency. If the switching hardware cannot act fast enough for successive note events, the robot may not be able to play the notes. This has already been a risk in the prototype design, as a MOSFET driver board was assumed to be available, but was actually a relay board. Testing showed that relay hardware could not act fast enough for certain passages. This risk will be mitigated in the ongoing design by the use of MOSFET switches, PWM control, and a stepwise approach from a single key to a small cluster before integration.

A second major technical risk is that there may be misalignment between the solenoid plungers and the keys. This could cause noise, inconsistency, and side loading. This risk can be mitigated through the use of adjustable mounting arms, brackets, contact pads, and a stepwise approach from a single key before integration.

A third major technical risk here is that there may be a tendency for the design to get too complex for a capstone project to build and verify in a timely fashion. While a full keyboard, elaborate exterior design, and sustain pedal features would be nice additions, there may be a tendency for these features to cloud the main design goals if pursued too aggressively. Therefore, the design will focus on a subset for the capstone while allowing for expansion in the future after reliable functionality has been demonstrated.

2.3 Assumptions

It is assumed that a home keyboard will be available that has sufficient key geometry for mounting, that solenoids can be obtained that have sufficient force and actuation time for the design, that framing materials can be reused without safety issues, and that the software design can be organized so that MIDI note events can be converted to Arduino commands without

human intervention during a song run.

3. Design Process and Alternatives

The concept was developed as a result of a structured design process as opposed to personal preference. The steps taken to develop the concept included translating requirements into specifications, building proof-of-concept hardware to identify key risks, and developing multiple system-level concepts and then selecting based on weighted criteria to match customer requirements.

3.1 Requirements Translation and Benchmarking

The starting point in the design process was to take customer requirements like portability, pleasant sound, and variety in notes and translate these into requirements that would drive the design. Benchmarks were also considered to identify how to make it successful in a public display environment. The requirements translated into how fast it should take to set up, how engaging the mechanism should be, and how reliable it should function so that users are not discouraged by failures. It was also realized at this stage that it was an integrated mechatronics problem involving mounting, electrical switching, timing, and user interface.

3.2 Proof-of-Concept Development

The development of a proof of concept was a key activity in the design process. The initial proof of concept was developed by placing a cardboard frame over a keyboard. It was able to play notes and even a rough version of “Mary Had a Little Lamb.” It was realized at this stage that there were significant integration challenges. The initially proposed hardware was not appropriate for the tempo required in playing. The steps taken from this point in the design process were rational and risk retirement-based. These steps included drawing a wiring diagram for one solenoid, designing a solenoid mount and interface to the keyboard, building a housing for MOSFET, Arduino, PWM, and voltage splitting, fixing bugs in the MIDI reading code, streamlining the Python to Arduino translation, and organizing the code to make it expandable to 61 keys.

3.3 Concept Generation

There were three main concepts that were analyzed, and they included the following:

For Concept A, the approach was based on the use of an array of solenoids, where each key in the range would have an actuator assigned to it [6].

For Concept B, the approach was to have fewer actuators moving over many keys via the use of a moving carriage or rail-based approach.

For Concept C, the approach was the use of linkage or cam-based multiplexing.

All the options had the capability of playing keys, but the issue was determining the best approach in terms of meeting the requirements of the project's priorities. Concept A offered the best approach in terms of the autonomous multi-note play, where each key had an actuator assigned to it without the need for the actuator moving in the lateral direction.

Concept B reduced actuator count, but introduced carriage motion and control complexity that increased schedule risk. For Concept C, the approach offered the advantages of fewer actuators, but the backlash, tuning, and lack of independent control of the playing of notes offered challenges in accurate note execution.

Based on the requirements of the customers, the key factors included the playing of the notes correctly, the sound being pleasing, autonomous, portable, and expandable, where the simplest approach in terms of the number of parts was not necessarily the best.

3.4 Downselect Criteria and Results

The downselect criteria were focused on the key aspects of the project, where the key factors included the autonomous approach, the range of notes, the dynamics, the non-invasive approach, the portability, the build complexity, the cost, and the expandability of the final product. The representative weighted decision matrix used for the downselect is shown in Table 2.

Based on the requirements, the final product had to have the capability of playing multiple notes, where the timing was crucial in terms of the public demonstrating the capabilities of the product, where the credibility of the product would only last for a short time if the playing of the notes was not done well.

Table 2. Representative weighted decision matrix for the major system concepts.

Criterion	Weight	Concept A: Fixed Array	Concept B: Moving Carriage	Concept C: Linkage/Cam
Autonomous reliability	0.25	5	2	2
Note range and dynamics potential	0.20	4	3	2
Non-invasive mounting and portability	0.15	4	3	3
Build complexity risk	0.15	3	2	2
Cost and fabrication burden	0.10	3	4	4
Future expandability	0.15	5	3	2
Weighted total	1.00	4.15	2.70	2.35

Concept A was given the highest weighted score since it was the most suitable solution to the project's most significant requirements, even though it was not the most cost-efficient solution. The main disadvantage of this solution is that it increases the number of channels, leading to increased wiring and power requirements. However, this disadvantage is feasible in a staged capstone project, while the other solutions compromise the project timeline and multi-note capability by reducing the number of hardware components used.

4. Proposed Design Solution

4.1 System Overview

The proposed design solution is a modular fixed array keyboard playing robot. The song data is preprocessed on a host computer and then transmitted to a Musical Bot Encore controller based on an Arduino board. The controller schedules the note times and sends PWM signals to the MOSFET driver electronics, which switch the current to the solenoids. The solenoids are mounted on adjustable arms connected to a non-invasive structure positioned above the keyboard. This design solution decouples the software timing, power switching, and actuation to enable independent subsystems to be improved without requiring changes to the entire system.

4.2 Mechanical Architecture

The proposed solution's mechanical architecture is based on a lightweight structural bridge that connects to a home keyboard without any need to permanently modify the keyboard. Adjustable arms or brackets hold the solenoids above the corresponding keyboard key to enable tuning of the positions during assembly. This design supports compatibility with multiple home-keyboard geometries while also enabling rapid setup and disassembly, allowing many noncritical support parts to be made from recycled parts.

Each of the actuation channels conforms to a uniform hardware arrangement, with a bracket, solenoid, plunger guide, and key contact interface. This modularity improves serviceability, as well as expansion, because the key station can be tested, adjusted, and then repeated across the range of keys or keyboard layouts the team elects to implement. Compliant contact pads minimize noise and local stress at the key interface, while adjustability helps maintain repeatable travel and ordered motion during demonstrations [7].

4.3 Electrical and Control Architecture

In the electrical architecture, the project utilizes the Arduino microcontroller to schedule the electrical signals. This microcontroller works with MOSFET electronics, which serve as the drivers of the electrical signals. This is necessary because the microcontroller sends the timing signals, but the solenoids need more current than the microcontroller's pins can provide. This change from the original relay hardware represents an important correction identified during prototype development. In the electrical architecture, the power supply is centralized, fused, and encased. This is necessary because the project will transport the keyboard to be used during demonstrations at public events. As the number of actuated keys increases, total current demand will also increase, so power distribution and integration testing will be critical during system expansion and public demonstrations.

4.4 Software Pipeline and Performance Strategy

The software pipeline involves the selection of the song, the processing of the MIDI events, and the eventual transfer of the events to the Arduino for playing the song. Some of the important tasks include fixing the bug in the script used for reading the MIDI, as well as the transfer of the Python script to the Arduino, which is crucial for the autonomous playing of the songs rather than housekeeping tasks. Using PWM is a realistic approach towards incorporating dynamics, although the expressive nuances are not necessarily important for the success of the program, the control over the energy of the strikes, as well as the duration, is important in ensuring that the playing does not appear flat or robotic over time [8]. The structure of the code, although long in the current form, is important in the long run for incorporating the keys as well as the sustain pedals, although they would have to follow the successful demonstration of the autonomous playing of the songs.

4.5 Advantages and Disadvantages Relative to Customer Requirements

The selected design has five advantages compared to the customer requirements. First, the design can be electronically controlled and operated autonomously. Second, the fixed-array design naturally supports overlapping notes compared to the moving or multiplex designs. Third, the non-invasive adjustable frame design supports the home keyboards. Fourth, the design can be built with substantial reused content in the structure while still maintaining a clear architecture for future teams to build upon. Fifth, the design can be visually comprehended for an audience at an Engineering Expo.

The design also has some disadvantages compared to the customer requirements. The design requires more channels, wires, and power management compared to the moving carriage and linkage designs. Additionally, there is more emphasis on mechanical alignment between key stations.

5. Go-Forward Plan and Verification

5.1 Why This Design Is the Best Choice

The selected design is the best choice for this project because it meets the highest-priority requirements simultaneously without relying on speculative coordination strategies. The design also supports autonomous operation, supports playing more than one note at a time compared to playing just a single note, supports non-invasive attachment to a home keyboard, and supports a modular design for staged development. The design also draws on experience from prototype development, such as the move from relay to MOSFET switching.

5.2 Implementation Plan

Implementation must occur in well-defined stages: Stage 1: Validate one full actuation channel; Stage 2: Expand to a small group of adjacent keys; Stage 3: Integrate demonstration subset, hardware, user interface; Stage 4: Focus on repeatability testing, optimization, exterior design, and full song testing.

Table 3. Near-term implementation work packages aligned with the team workflow.

Work Package	Purpose	Near-Term Output
Mechanical mounting design	Create adjustable, non-invasive key-actuation geometry	CAD for solenoid mounts and keyboard interface
Electrical integration	Replace relay-based switching with proper high-speed driver hardware	MOSFET/PWM housing, wiring diagram, and safe power layout
Software pipeline cleanup	Enable reliable autonomous playback from song input to controller output	Bug-fixed code, organized storage, and streamlined Python-to-Arduino handoff
Validation and analysis	Retire risk before full integration	Single-key tests, small-cluster tests, and stress-analysis support

5.3 Verification Against Engineering Specifications

Verification must be directly related to specifications listed in Table 1. Autonomous reliability must be verified for 20 full songs with a definition of success as completion without human input from start. The song library requirement is satisfied when at least three songs can be selected for playback on demand. Setup must be verified for deployment from transport to playback ready within a timed period. System mass must also be verified to be less than 100 lb. Material must be verified to be 50% reused by part count to avoid assumption based on mass.

The quality of performance should be assessed by event-based testing and complete piece demonstrations. At the channel level, the team should ensure proper triggering, consistency, and thermal behavior during repeated playback. At the system level, the team should ensure that the keyboard is producing the right notes, that the mechanism is visually orderly and safe, and that

the sound is pleasing to the ear for public consumption.

6. Conclusion

Musical Bot Encore is a keyboard-playing robot designed to replace an aging outreach exhibit with a more reliable and extensible Engineering Expo demonstration. The design process began with stakeholder requirements, specification of these requirements, proof of concept to identify key risks, and comparing various system concepts based on weightings related to project goals.

The proposed design is not the lowest channel count design; rather, it is the strongest capstone design based on the requirements most critical to project success: autonomous operation, recognizable multi-note play, portability, support for various home keyboard types, and a path to future expansion.

The implementation plan is designed to retire hardware, software, and integration risks early in the project plan. If successfully executed, Musical Bot Encore will deliver an engaging and technically defensible exhibit to Oregon State University for future Engineering Expo attendees to engage with.

References

- [1] Weinberg, G., and Driscoll, S., “The Design of a Robotic Marimba Player: Introducing Pitch into Robotic Musicianship,” *Proceedings of the Conference on New Interfaces for Musical Expression (NIME)*, New York, NY, 2007, pp. 228–233.
- [2] Hoffman, G., and Weinberg, G., “Interactive Improvisation with a Robotic Marimba Player,” *Autonomous Robots*, Vol. 31, Nos. 2–3, 2011, pp. 133–153.
- [3] Grindlay, G., and Helmbold, D., “Modeling, Analyzing, and Synthesizing Expressive Piano Performances with Graphical Models,” *Machine Learning*, Vol. 65, Nos. 2–3, 2006, pp. 361–387.
- [4] Demos, A. P., Lisboa, T., and Chaffin, R., “Flexibility of Expressive Timing in Repeated Musical Performances,” *Frontiers in Psychology*, Vol. 7, 2016, Art. No. 1490.
- [5] Repp, B. H., “Variability of Timing in Expressive Piano Performance Increases with Interval Duration,” *Psychonomic Bulletin & Review*, Vol. 4, No. 4, 1997, pp. 530–534.
- [6] Muramatsu, S., Fujiwara, Y., Hayashida, H., and Yamamoto, J., “Solenoid Actuator for Use in Automatic Performance Piano,” US4513652A, Apr. 30, 1985.
- [7] Acker, A., and Birkett, S., “Recent Research in Piano Key Action,” presented at Haptics 2009, Rochester, NY, Feb. 22–24, 2009.
- [8] Fields, K. D., Mclam, G. E., and Yorba, A. J., “Method and Apparatus for Actuating Solenoids in a Player Piano,” EP0922276A1, June 16, 1999.

Artifact 4: Individual Self-Learning Assignment

Required item: Individual self-learning assignment

Included file: derek-martin-self-learning-assignment.pdf

Fusion 360 self-learning report applied to the robotic piano player.

WIC Self-Learning Report

Learning the Basics of Fusion 360 for a Solenoid-Driven Robotic Piano Player

Student: Derek Martin

Project: Robotic piano player using solenoids

Assignment alignment: Mechanical and Manufacturing Engineering Topic (3) - Learn the basics of Fusion 360

Date: April 13, 2026

This report documents a self-directed learning activity that most closely matches the course example "Learn the basics of Fusion 360." The training was completed through YouTube and Autodesk Fusion tutorials, then applied to the capstone design process.

Self-Learning Report

Capstone project	Autonomous robotic piano player that uses solenoids to actuate piano keys
Skill learned	Basics of Autodesk Fusion 360 for 3D modeling and design iteration
Training time	Approximately 7 hours of self-directed instruction before additional practice and project use
Primary sources	Four Product Design Online YouTube lessons plus Autodesk Fusion tutorials and help resources
Project impact	Supported frame design, design iteration, and stress/loading considerations related to the solenoids and frame

1. Skill Learned and Why I Chose It

The self-directed learning topic that I have selected is the basics of Autodesk Fusion 360. Although there are multiple examples in the class syllabus, this example aligns most with "Learn the basics of Fusion 360" since the learning occurred via YouTube tutorials and Autodesk help files rather than LinkedIn Learning. Prior to undertaking this task, my experience in 3D modeling was limited, which meant that Fusion 360 remained unfamiliar to me in practical terms as an engineer. At best, I knew what computer-aided design was capable of doing, though I did not yet have sufficient proficiency in it to make parts quickly and efficiently.

Fusion 360 was selected because I am designing a capstone robot that plays a piano using solenoids; it involves making numerous design choices regarding geometry and frame mounting. It became necessary to model various designs quickly, evaluate different design choices visually, and communicate them with others. As such, learning this software would enable me to acquire some attributes in design that will come in handy during the capstone process, such as spatial reasoning, digital prototyping, design communication, iterative design thinking, and relating geometry to the loading and manufacturing considerations.

2. Self-Learning Training and Evidence

Over the period of seven hours, I taught myself how to use Autodesk Fusion 360 by going through video tutorials and training materials published by Autodesk. The primary path in training I took was the beginner level Product Design Online sequence of tutorials. Instead of following the video lessons passively, I worked inside Fusion 360 on each step shown. This helped me familiarize myself with interface navigation, sketching techniques, feature tree, and difference between designing the model and conceptualizing the object. I additionally went through Autodesk Fusion 360 tutorials, which helped me get accustomed with the fundamentals of the software.

Each tutorial had more advanced models for practice than the previous one. Firstly, I learned sketching, extrude, shell, and pattern features by modeling a Lego handle-like object. Then, I learned how to include a picture into the design, use splines, and create a revolve body by modeling a glass bottle. Then, by modeling a paper clip, I familiarized myself with sweep and tangent arc tool functions. Lastly, I modeled another bottle with the help of lofts and construction planes. These

practice models are depicted in Fig. 1. These documents are proof that I did not merely watch tutorial videos but designed these models.

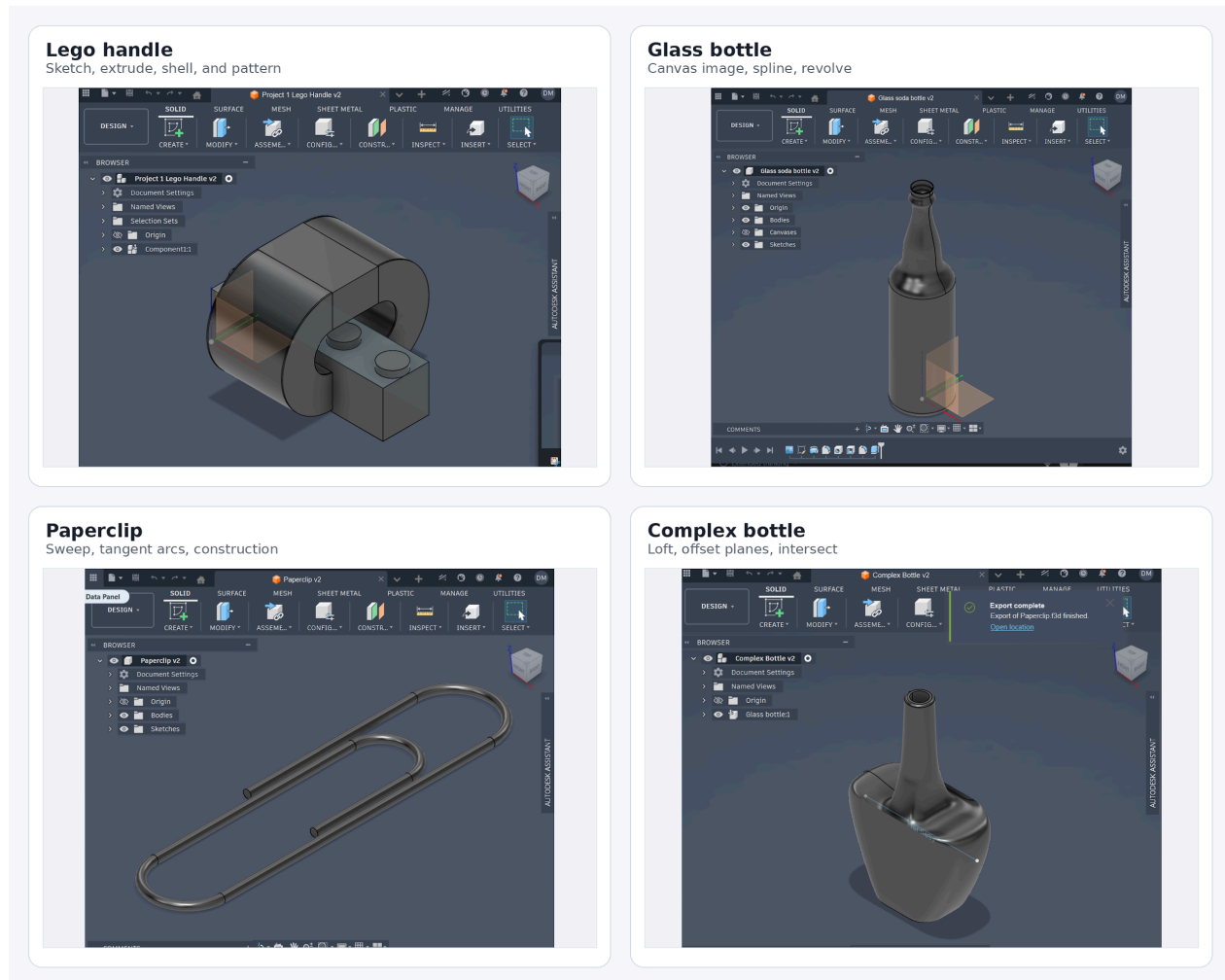


Figure 1. Practice models completed during self-learning in Fusion 360: Lego handle, glass bottle, paperclip, and complex bottle.

3. Application to the Capstone Project

Following the guided practice, I was able to use the same process on the robotic piano player. In this particular project, Fusion 360 was helpful in allowing me to have an easy way to construct and modify the frame digitally as compared to relying solely on trial and error to achieve success. It helped us to model the geometry of the frame digitally, helped us keep track of the design as it evolved, and assisted in thinking through the loading process for the solenoids. In this project, that is especially important since any change in geometry could impact alignment, package size, stiffness, and ease of assembly. Figure 2 shows the Fusion 360 assembly model and drawing package that I developed for the robotic piano player. These project files allowed me to evaluate the frame layout, the placement of the solenoid bank, and the overall assembly dimensions before fabrication.

The greatest change in ability came in terms of efficiency. Once I had mastered the basics of the program, I could go from conception to model in seconds, something I never could do before. This meant that instead of guessing and testing the shape and size of each component until finding an

acceptable configuration, I could design it virtually first and modify it accordingly. Not only was this process more efficient as far as building and testing was concerned, but it was beneficial in improving communication in the project overall, as models are easier to talk about than ideas.

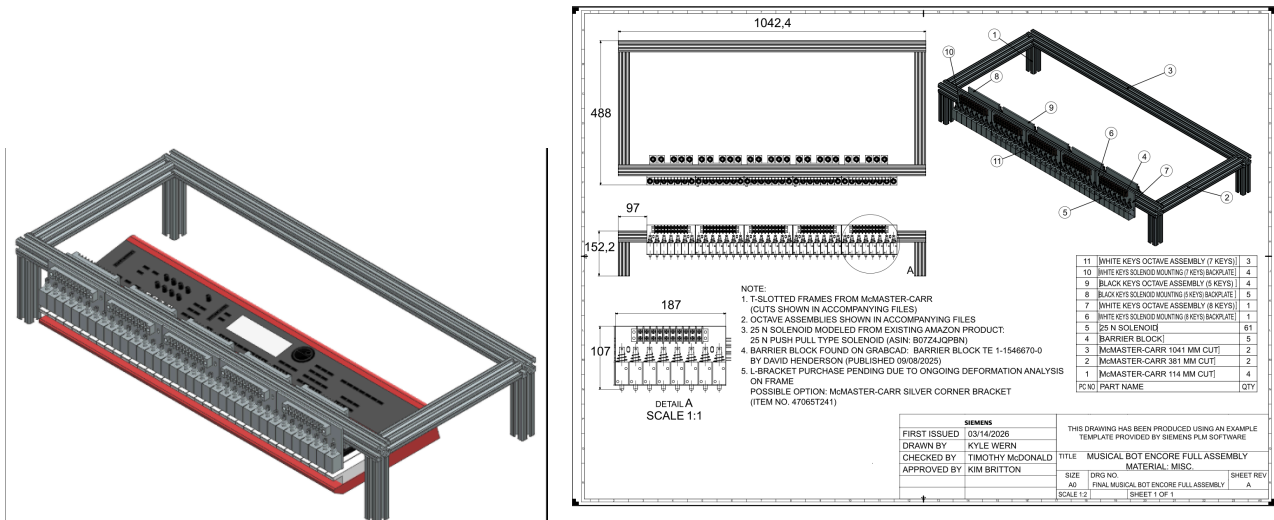


Figure 2. Fusion 360 application to the capstone project: (left) CAD assembly of the robotic piano player showing the frame, keyboard support structure, and solenoid mounting layout; (right) engineering drawing generated from the model showing dimensions, part relationships, and assembly details used for design review and refinement.

4. Results and Design Attributes Developed

The primary outcome of the learning process has been making Fusion 360 a viable software tool to employ on my capstone project as opposed to barely knowing what it looked like or being capable of launching it properly. As shown in Fig. 2, the new Fusion 360 skills were applied to both 3D assembly modeling and technical drawing creation, which made the design easier to evaluate, communicate, and refine before building. This allowed for designing faster while eliminating any unnecessary assumptions and providing better grounds to iterate prior to fabricating or testing anything. Moreover, this process helped me adopt certain mindsets regarding the design process, such as focusing on proper part assembly, load transfer through a structure, and effective changes in case the concept should evolve. All of those skills could immediately apply in practice.

Overall, I have managed to improve my competencies in 3D modeling, making decisions based on my designs, iteration, and linking my choices with performance considerations. These skills are particularly important for a solenoid-powered piano player since there would be a lot of iterations involved in order to develop a viable concept. Mastering Fusion 360 will undoubtedly help me perform better at my future capstone and in any position related to modeling and design.

References

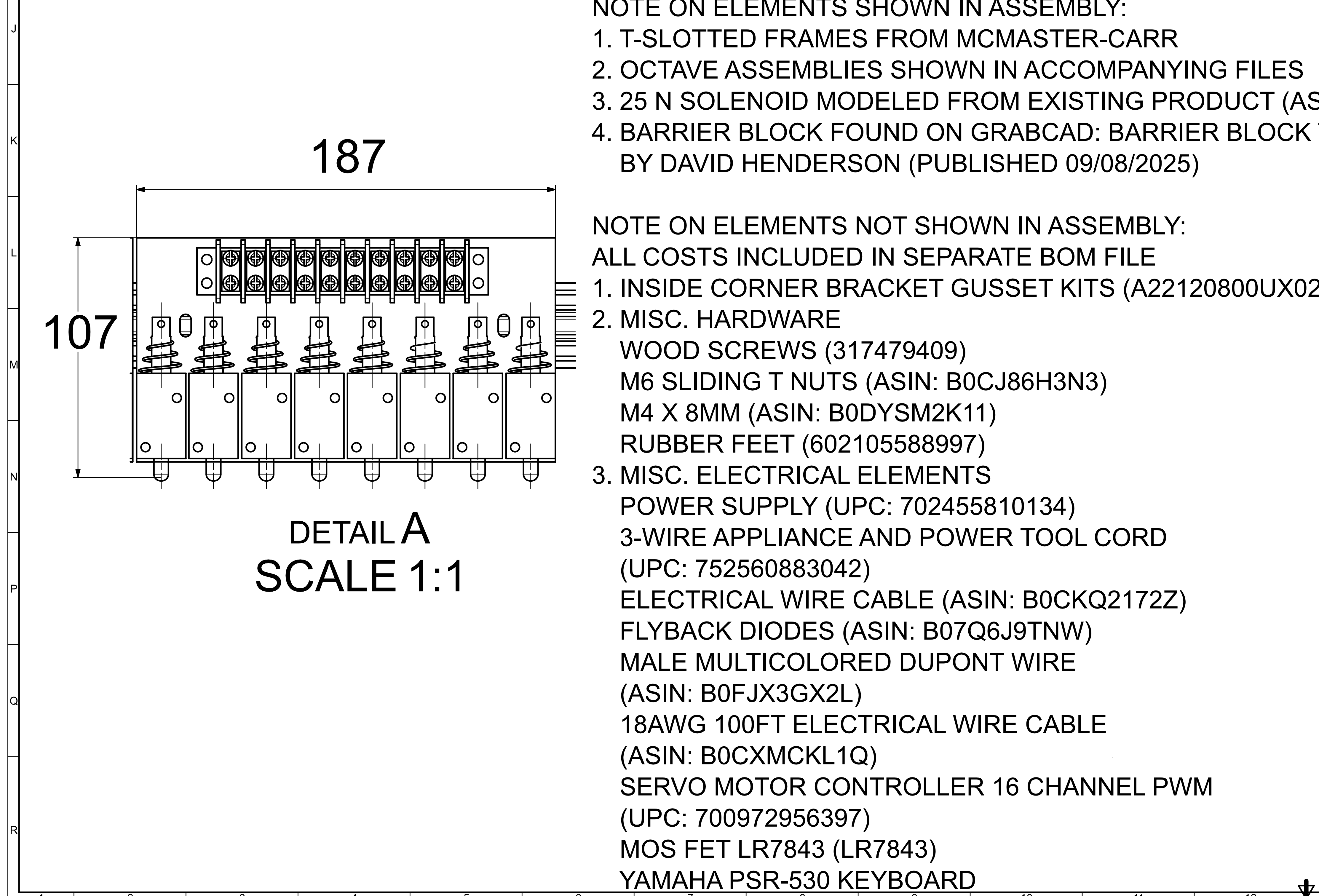
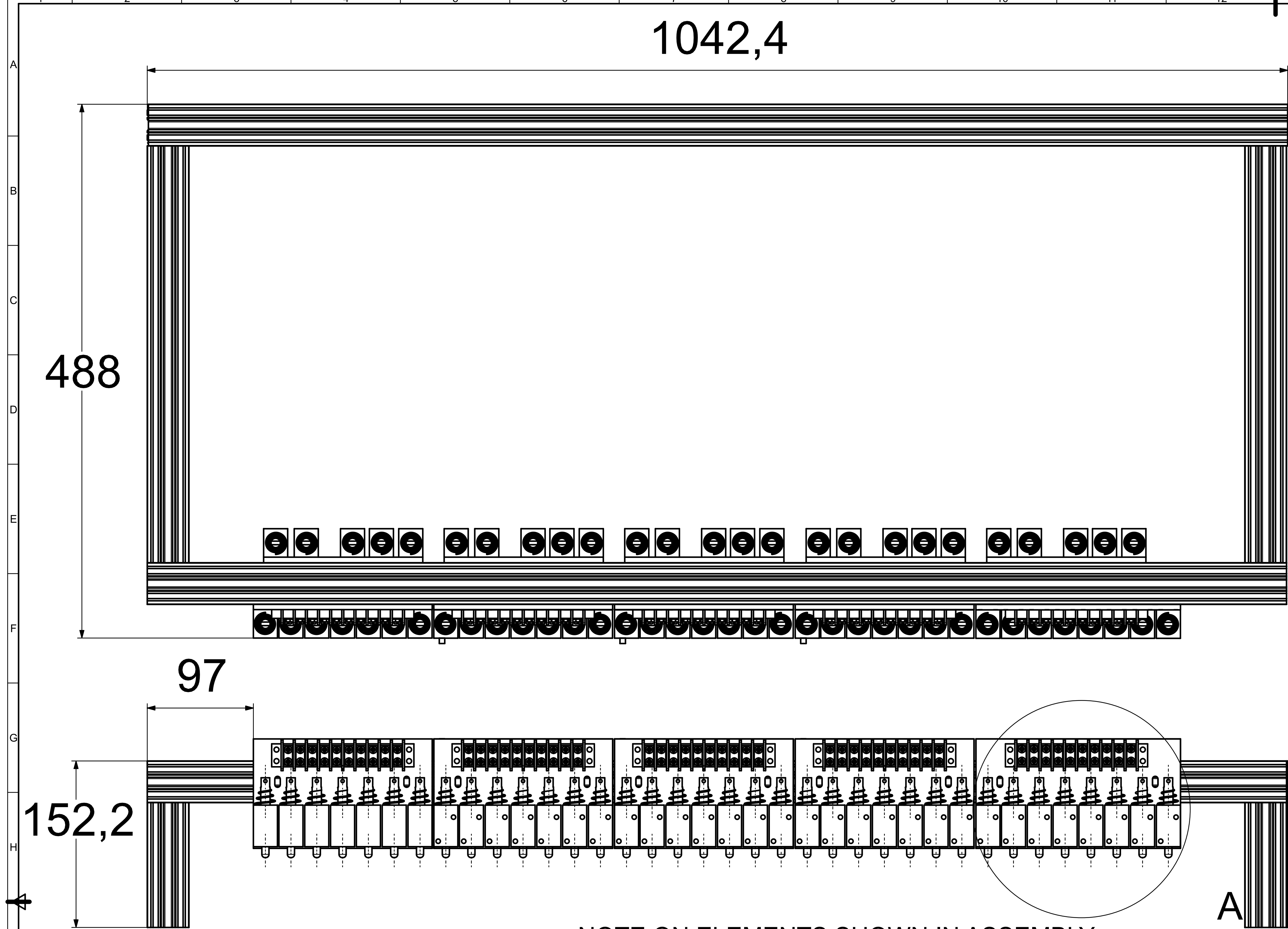
- [1] Product Design Online, "Day 1 of Learn Fusion 360 in 30 Days for Complete Beginners! - 2023 Edition," YouTube. [Online]. Available: <https://www.youtube.com/watch?v=d3qGQ2utl2A>. [Accessed: Apr. 13, 2026].
- [2] Product Design Online, "Day 2 of Learn Fusion 360 in 30 Days for Complete Beginners! - 2023 Edition," YouTube. [Online]. Available: <https://www.youtube.com/watch?v=DfAfxae8aRc>. [Accessed: Apr. 13, 2026].
- [3] Product Design Online, "Day 3 of Learn Fusion 360 in 30 Days for Complete Beginners! - 2023 Edition," YouTube. [Online]. Available: <https://www.youtube.com/watch?v=ZReXmjT42DM>. [Accessed: Apr. 13, 2026].
- [4] Product Design Online, "Day 4 of Learn Fusion 360 in 30 Days for Complete Beginners! - 2023 Edition," YouTube. [Online]. Available: <https://www.youtube.com/watch?v=9Y-3W-Pr64U>. [Accessed: Apr. 13, 2026].
- [5] Autodesk, "Self-paced learning for Fusion." [Online]. Available: <https://www.autodesk.com/learn/ondemand/collection/self-paced-learning-for-fusion>. [Accessed: Apr. 13, 2026].
- [6] Autodesk, "Autodesk Fusion tutorials for beginners and students." [Online]. Available: <https://www.autodesk.com/solutions/design-manufacturing/fusion-tutorials>. [Accessed: Apr. 13, 2026].

Artifact 5: Final Team Report

Required item: Final team report

Included file: team044-pianobot-final-drawing-package.pdf

Team 044 PianoBot final drawing package from the final submission bundle.



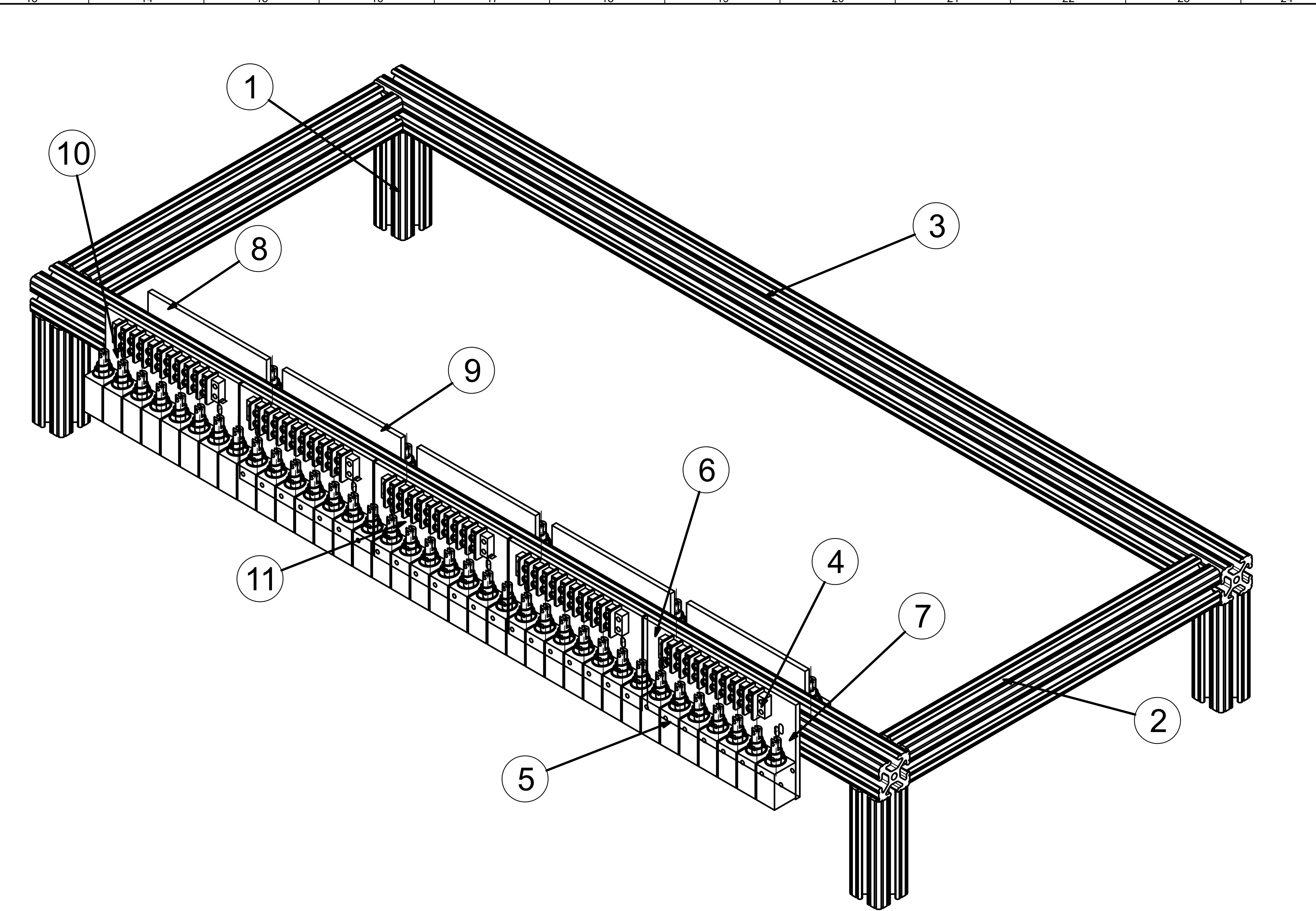
NOTE ON ELEMENTS SHOWN IN ASSEMBLY:

- 1. T-SLOTTED FRAMES FROM MCMMASTER-CARR
- 2. OCTAVE ASSEMBLIES SHOWN IN ACCOMPANYING FILES
- 3. 25 N SOLENOID MODELED FROM EXISTING PRODUCT (ASIN: B07Z4JQPN)
- 4. BARRIER BLOCK FOUND ON GRABCAD: BARRIER BLOCK TE 1-1546670-0 BY DAVID HENDERSON (PUBLISHED 09/08/2025)

NOTE ON ELEMENTS NOT SHOWN IN ASSEMBLY:

ALL COSTS INCLUDED IN SEPARATE BOM FILE

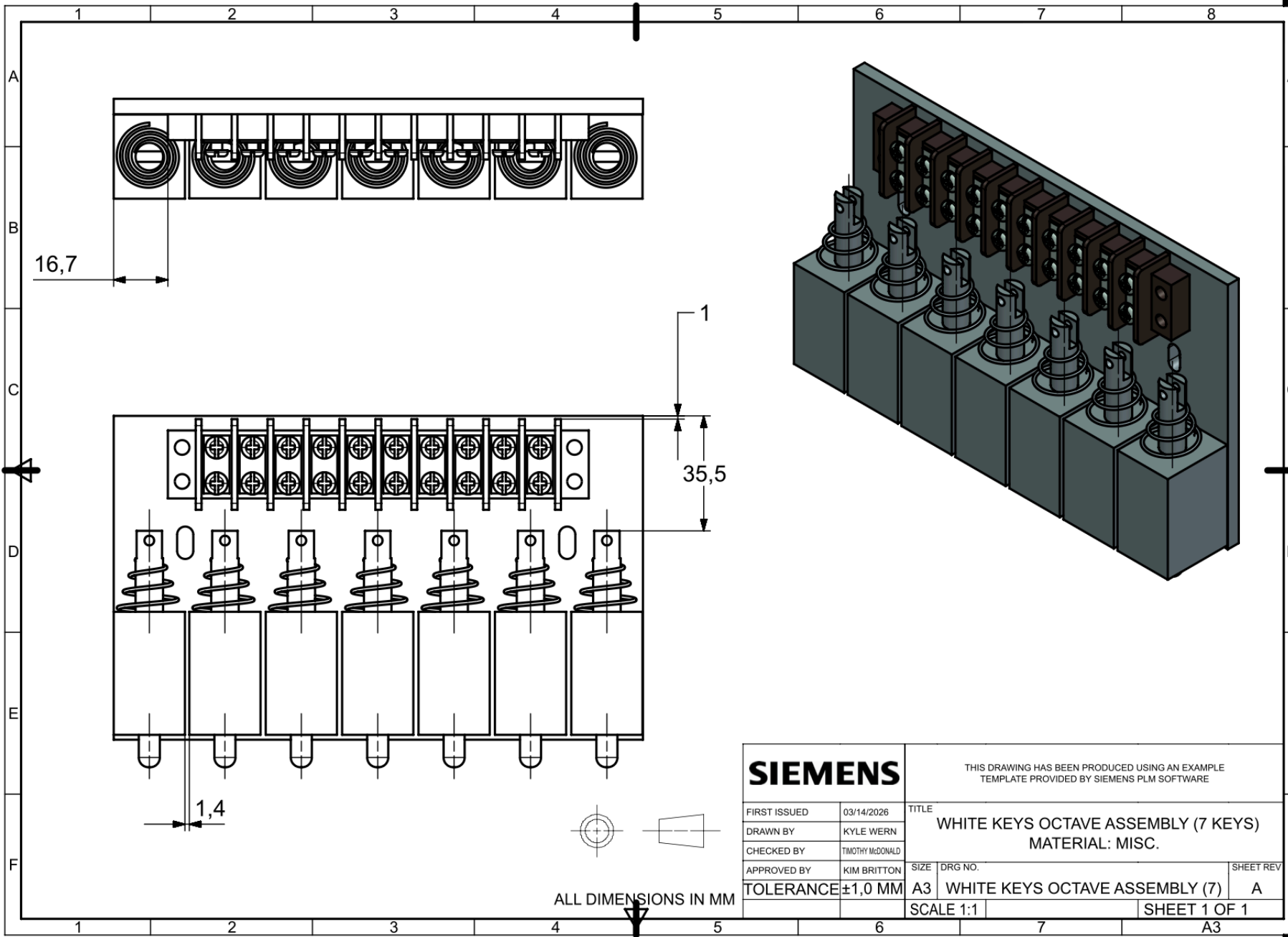
- 1. INSIDE CORNER BRACKET GUSSET KITS (A22120800UX0241)
- 2. MISC. HARDWARE
 - WOOD SCREWS (317479409)
 - M6 SLIDING T NUTS (ASIN: B0CJ86H3N3)
 - M4 X 8MM (ASIN: B0DYSM2K11)
 - RUBBER FEET (602105588997)
- 3. MISC. ELECTRICAL ELEMENTS
 - POWER SUPPLY (UPC: 702455810134)
 - 3-WIRE APPLIANCE AND POWER TOOL CORD (UPC: 752560883042)
 - ELECTRICAL WIRE CABLE (ASIN: B0CKQ2172Z)
 - FLYBACK DIODES (ASIN: B07Q6J9TNW)
 - MALE MULTICOLORED DUPONT WIRE (ASIN: B0FJX3GX2L)
 - 18AWG 100FT ELECTRICAL WIRE CABLE (ASIN: B0CXMCKL1Q)
 - SERVO MOTOR CONTROLLER 16 CHANNEL PWM (UPC: 700972956397)
 - MOS FET LR7843 (LR7843)
 - YAMAHA PSR-530 KEYBOARD



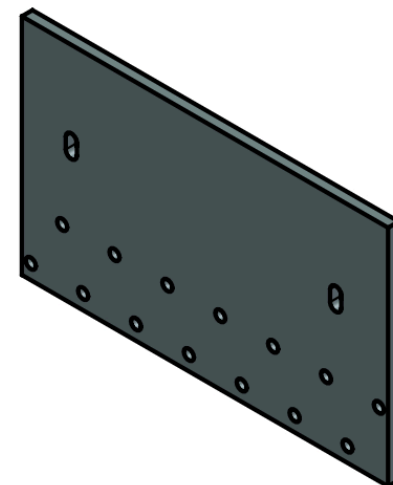
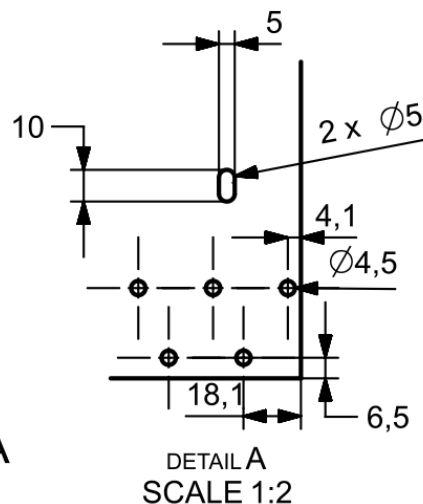
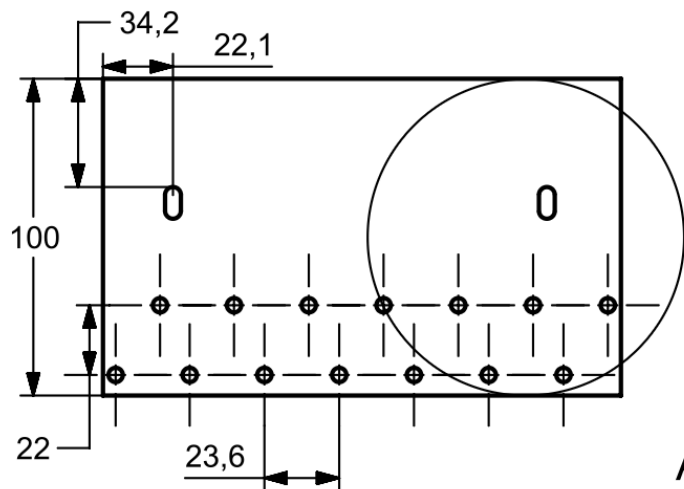
[N/A]	[MOSFET MOUNTS]	[4]	4.00
[N/A]	[PEDAL HOUSING ASSEMBLY]	[1]	1.00
PC NO	PART NAME	QTY	USD.

[N/A]	[PCA9685 MOUNTS]	[16]	16.00
11	[WHITE KEYS OCTAVE ASSEMBLY (7 KEYS)]	3	3.00
10	[WHITE KEYS SOLENOID MOUNTING (7 KEYS) BACKPLATE]	4	4.00
9	[BLACK KEYS OCTAVE ASSEMBLY (5 KEYS)]	4	4.00
8	[BLACK KEYS SOLENOID MOUNTING (5 KEYS) BACKPLATE]	5	5.00
7	[WHITE KEYS OCTAVE ASSEMBLY (8 KEYS)]	1	1.00
6	[WHITE KEYS SOLENOID MOUNTING (8 KEYS) BACKPLATE]	1	1.00
5	[25 N SOLENOID]	[62]	210.8
4	[BARRIER BLOCK]	5	27.78
3	[McMASTER-CARR 1041 MM CUT]	2	63.27
2	[McMASTER-CARR 381 MM CUT]	2	23.15
1	[McMASTER-CARR 114 MM CUT]	4	17.50
PC NO	PART NAME	QTY	USD.

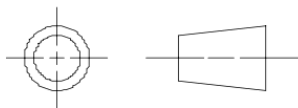
SIEMENS		THIS DRAWING HAS BEEN PRODUCED USING AN EXAMPLE TEMPLATE PROVIDED BY SIEMENS PLM SOFTWARE		
FIRST ISSUED	06/02/2026			
DRAWN BY	K. BRITTON	TITLE MUSICAL BOT ENCORE FULL ASSEMBLY MATERIAL: MISC.		
CHECKED BY	T. McDONALD			
APPROVED BY	K. WERN	SIZE	DRG NO.	SHEET REV
		A0	FINAL MUSICAL BOT ENCORE FULL ASSEMBLY	A
		SCALE 1:2	SHEET 1 OF 1	



SIEMENS		THIS DRAWING HAS BEEN PRODUCED USING AN EXAMPLE TEMPLATE PROVIDED BY SIEMENS PLM SOFTWARE		
FIRST ISSUED	03/14/2026	TITLE		
DRAWN BY	KYLE WERN	WHITE KEYS OCTAVE ASSEMBLY (7 KEYS)		
CHECKED BY	TIMOTHY McDONALD	MATERIAL: MISC.		
APPROVED BY	KIM BRITTON	SIZE	DRG NO.	SHEET REV
TOLERANCE	±1,0 MM	A3	WHITE KEYS OCTAVE ASSEMBLY (7)	A
		SCALE 1:1	SHEET 1 OF 1	



NOTE:
 2 X THRU SLOT 10 LG X 10 W X R5
 14 X 4,5 MM THRU HOLES
 1 MM TOLERANCE APPLIED
 FOR PETG FILAMENT PRINTING



ALL DIMENSIONS IN MM

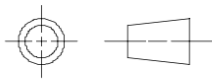
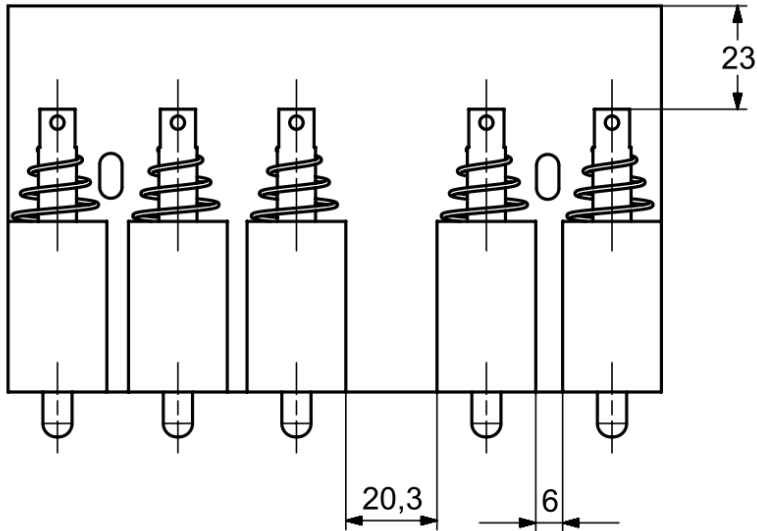
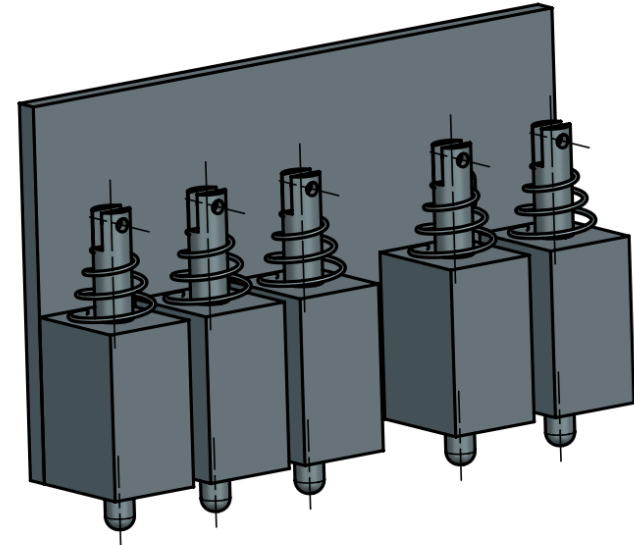
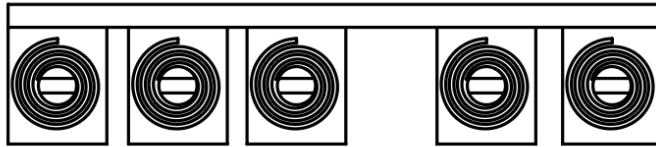
SIEMENS

THIS DRAWING HAS BEEN PRODUCED USING AN EXAMPLE
 TEMPLATE PROVIDED BY SIEMENS PLM SOFTWARE

FIRST ISSUED	03/14/2026
DRAWN BY	KYLE WERN
CHECKED BY	TIMOTHY McDONALD
APPROVED BY	KIM BRITTON
TOLERANCE	±1,0 MM

TITLE			WHITE KEYS SOLENOID MOUNTING (7 KEYS) BACKPLATE MATERIAL: PETG FILAMENT		
SIZE	DRG NO.				SHEET REV
A4	WHITE KEYS SOLENOID MOUNTING (7 KEYS) BACKPLATE				A
SCALE 1:1				SHEET 1 OF 1	

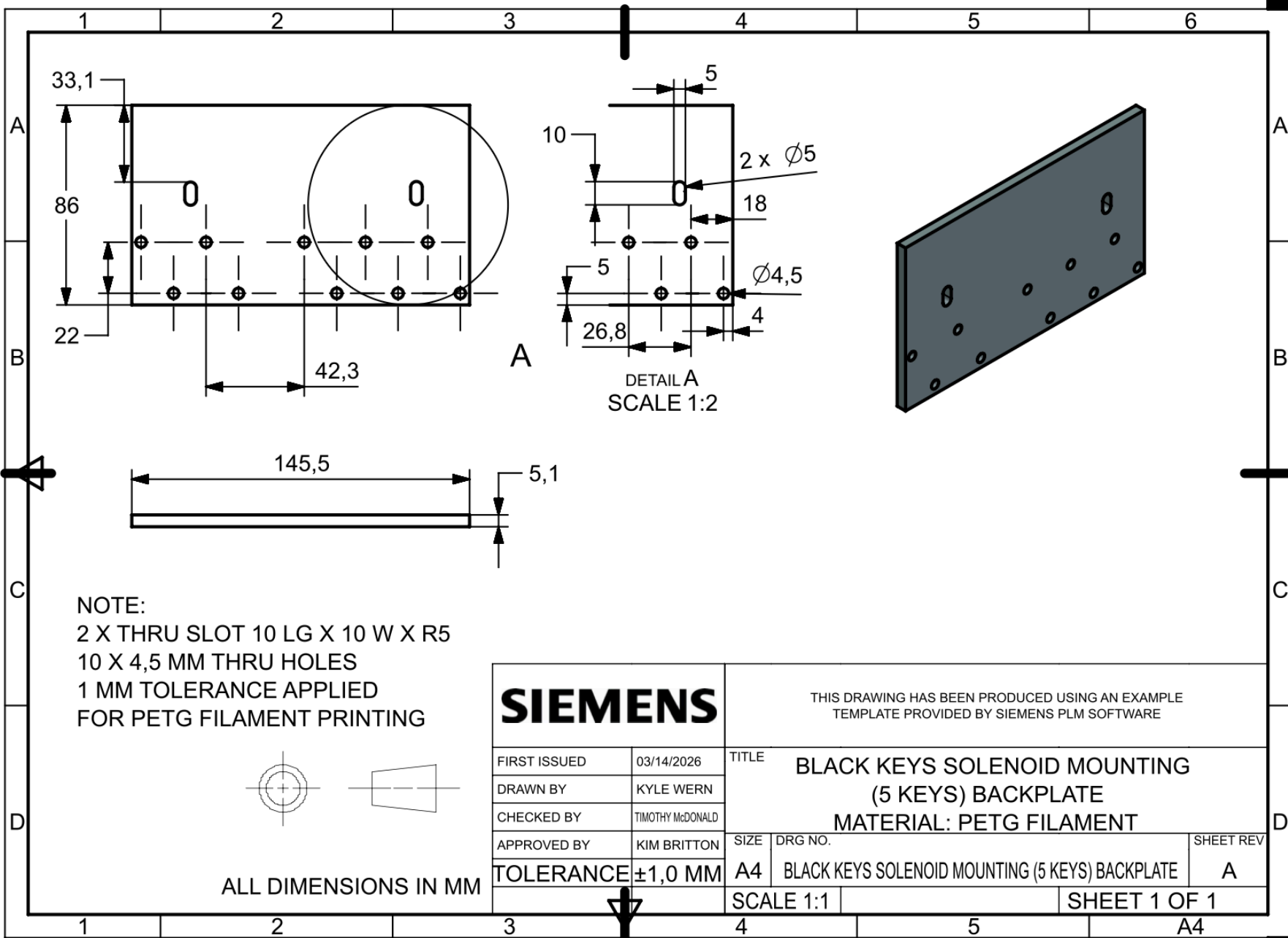
A4

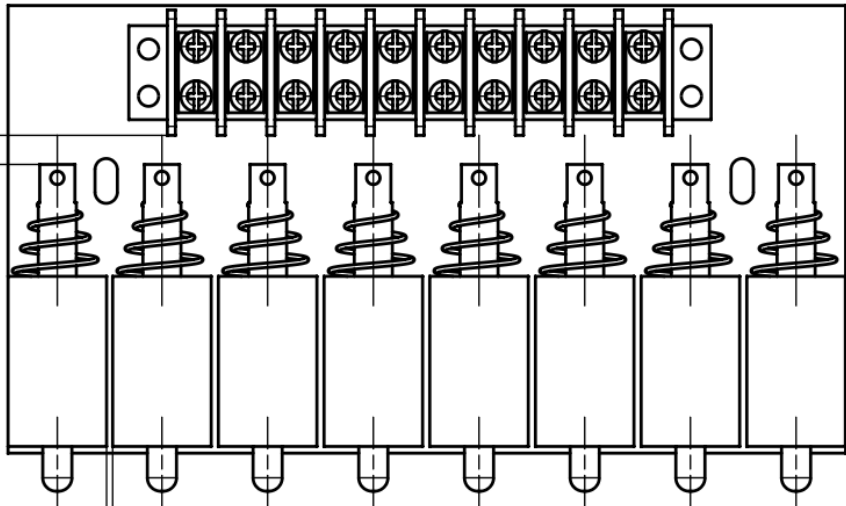
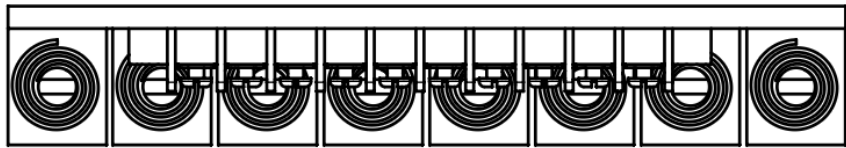


ALL DIMENSIONS IN MM

SIEMENS		THIS DRAWING HAS BEEN PRODUCED USING AN EXAMPLE TEMPLATE PROVIDED BY SIEMENS PLM SOFTWARE	
FIRST ISSUED	03/14/2026	TITLE	
DRAWN BY	KYLE WERN	BLACK KEYS OCTAVE ASSEMBLY (5 KEYS)	
CHECKED BY	TIMOTHY McDONALD	MATERIAL: MISC	
APPROVED BY	KIM BRITTON	SIZE	DRG NO.
TOLERANCE	±1,0 MM	A3	BLACK KEYS OCTAVE ASSEMBLY
		SCALE 1:1	SHEET 1 OF 1

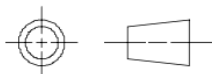
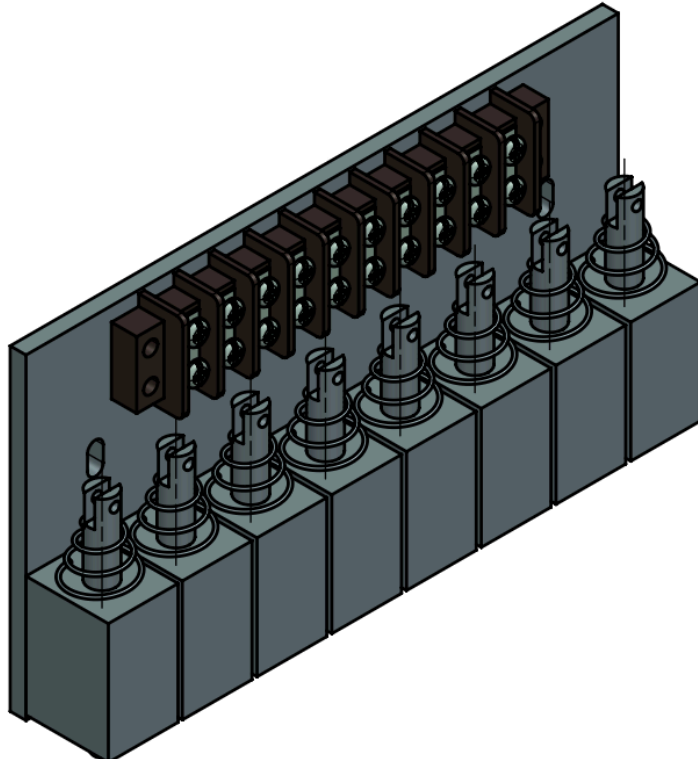
A3





6,5

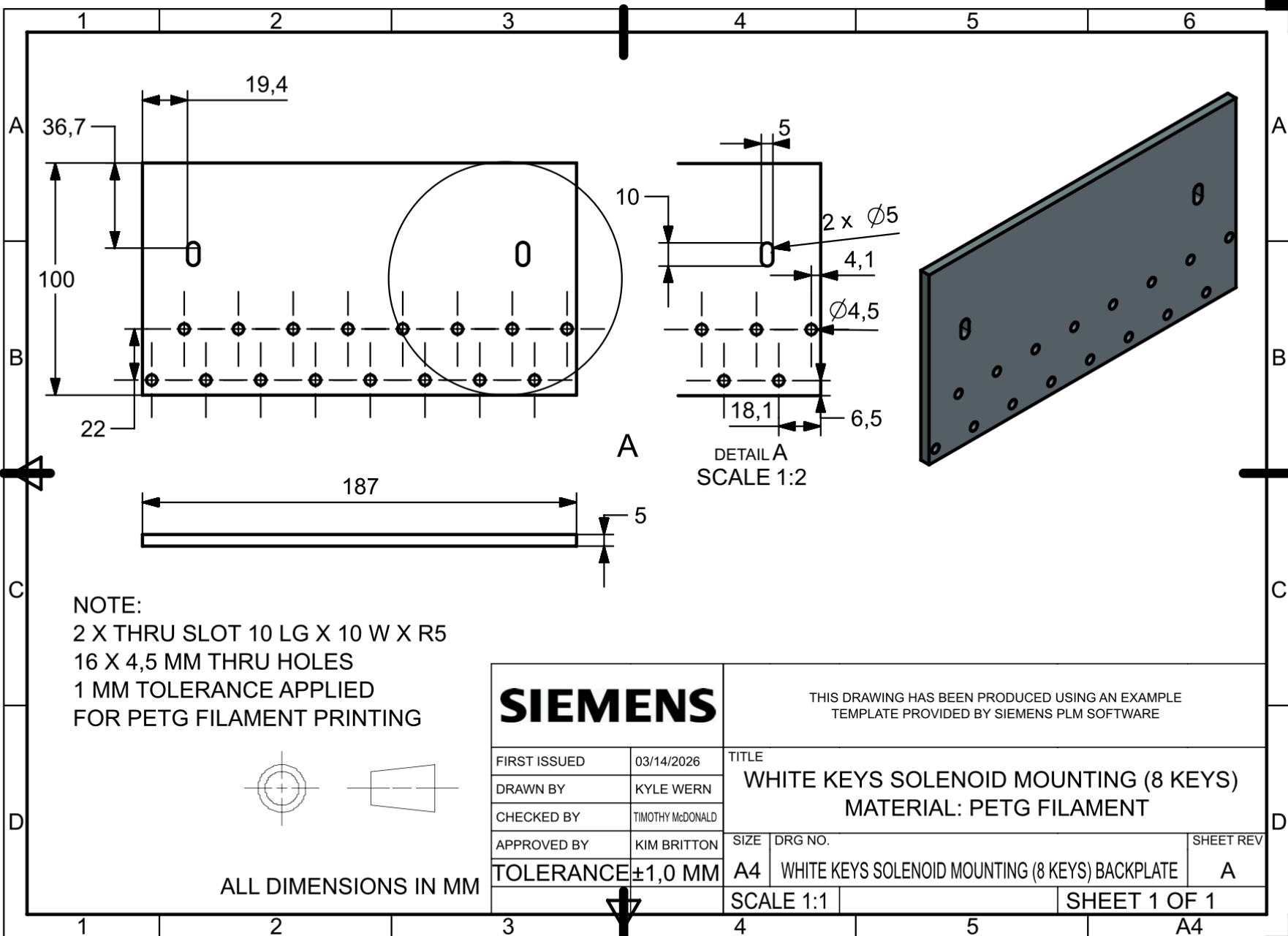
1,4

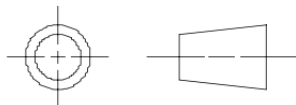
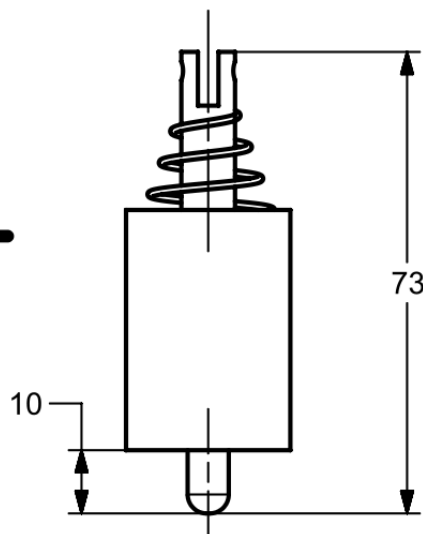
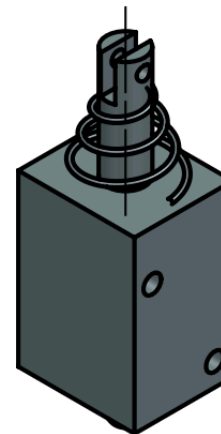
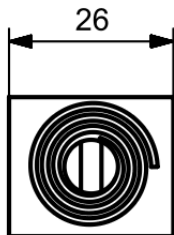


ALL DIMENSIONS IN MM

SIEMENS		THIS DRAWING HAS BEEN PRODUCED USING AN EXAMPLE TEMPLATE PROVIDED BY SIEMENS PLM SOFTWARE		
FIRST ISSUED	03/14/2026	TITLE		
DRAWN BY	KYLE WERN	WHITE KEYS OCTAVE ASSEMBLY (8 KEYS)		
CHECKED BY	TIMOTHY McDONALD	MATERIAL: MISC.		
APPROVED BY	KIM BRITTON	SIZE	DRG NO.	SHEET REV
TOLERANCE	±1,0 MM	A3	WHITE KEYS OCTAVE ASSEMBLY (8)	A
		SCALE 1:1	SHEET 1 OF 1	

A3





NOTE:
25 N SOLENOID MODELED FROM EXISTING AMAZON PRODUCT:
25 N PUSH PULL TYPE SOLENOID (ASIN: B07Z4JQPBN)

SIEMENS

THIS DRAWING HAS BEEN PRODUCED USING AN EXAMPLE
TEMPLATE PROVIDED BY SIEMENS PLM SOFTWARE

FIRST ISSUED	03/14/2026
DRAWN BY	KYLE WERN
CHECKED BY	TIMOTHY McDONALD
APPROVED BY	KIM BRITTON
TOLERANCE	N/A

TITLE 25 N PUSH PULL TYPE SOLENOID (ASIN: B07Z4JQPBN) MATERIAL: MISC.			
SIZE	DRG NO.	SHEET REV	
A4	25 N SOLENOID	A	
SCALE 1:1		SHEET 1 OF 1	

ALL DIMENSIONS IN MM

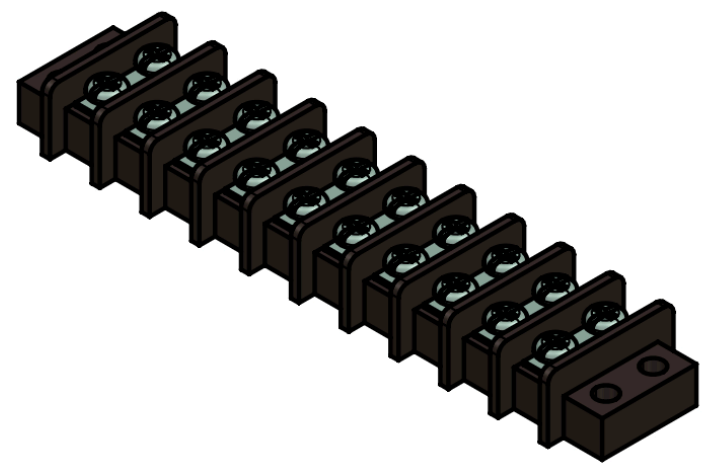
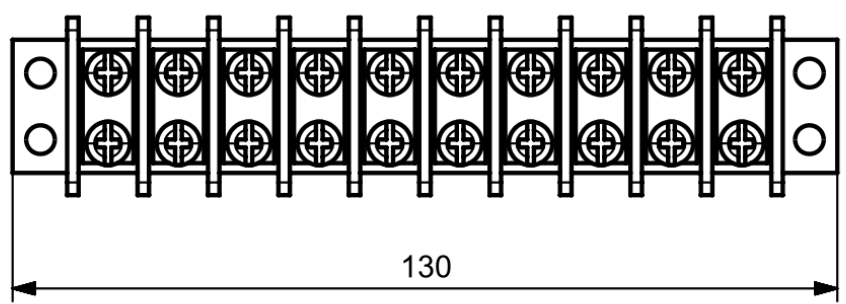
1 2 3 4 5 6

A

B

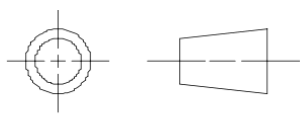
C

D



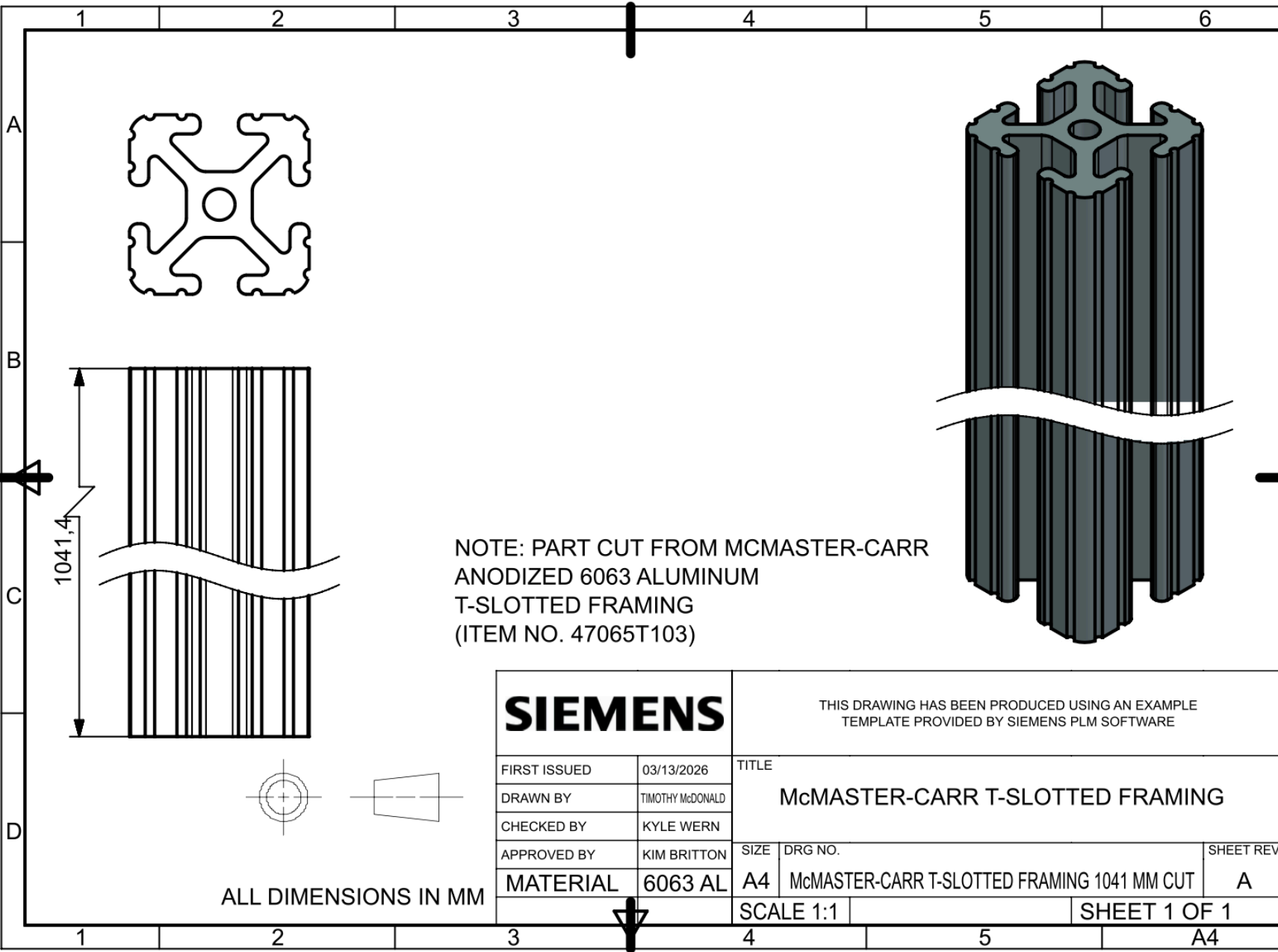
NOTE:
BARRIER BLOCK FOUND ON GRABCAD:
BARRIER BLOCK TE 1-1546670-0
BY DAVID HENDERSON (PUBLISHED 09/08/2025)

SIEMENS		THIS DRAWING HAS BEEN PRODUCED USING AN EXAMPLE TEMPLATE PROVIDED BY SIEMENS PLM SOFTWARE		
FIRST ISSUED	09/08/2025	TITLE BARRIER BLOCK TE 1-1546670-0 MATERIAL: MISC.		
DRAWN BY	DAVID HENDERSON			
CHECKED BY	N/A			
APPROVED BY	N/A	SIZE	DRG NO.	SHEET REV
TOLERANCE	N/A	A4	BARRIER BLOCK TE 1-1546670-0	A
		SCALE 1:1	SHEET 1 OF 1	



ALL DIMENSIONS IN MM

1 2 3 4 5 A4



NOTE: PART CUT FROM MCMaster-CARR
ANODIZED 6063 ALUMINUM
T-SLOTTED FRAMING
(ITEM NO. 47065T103)

SIEMENS

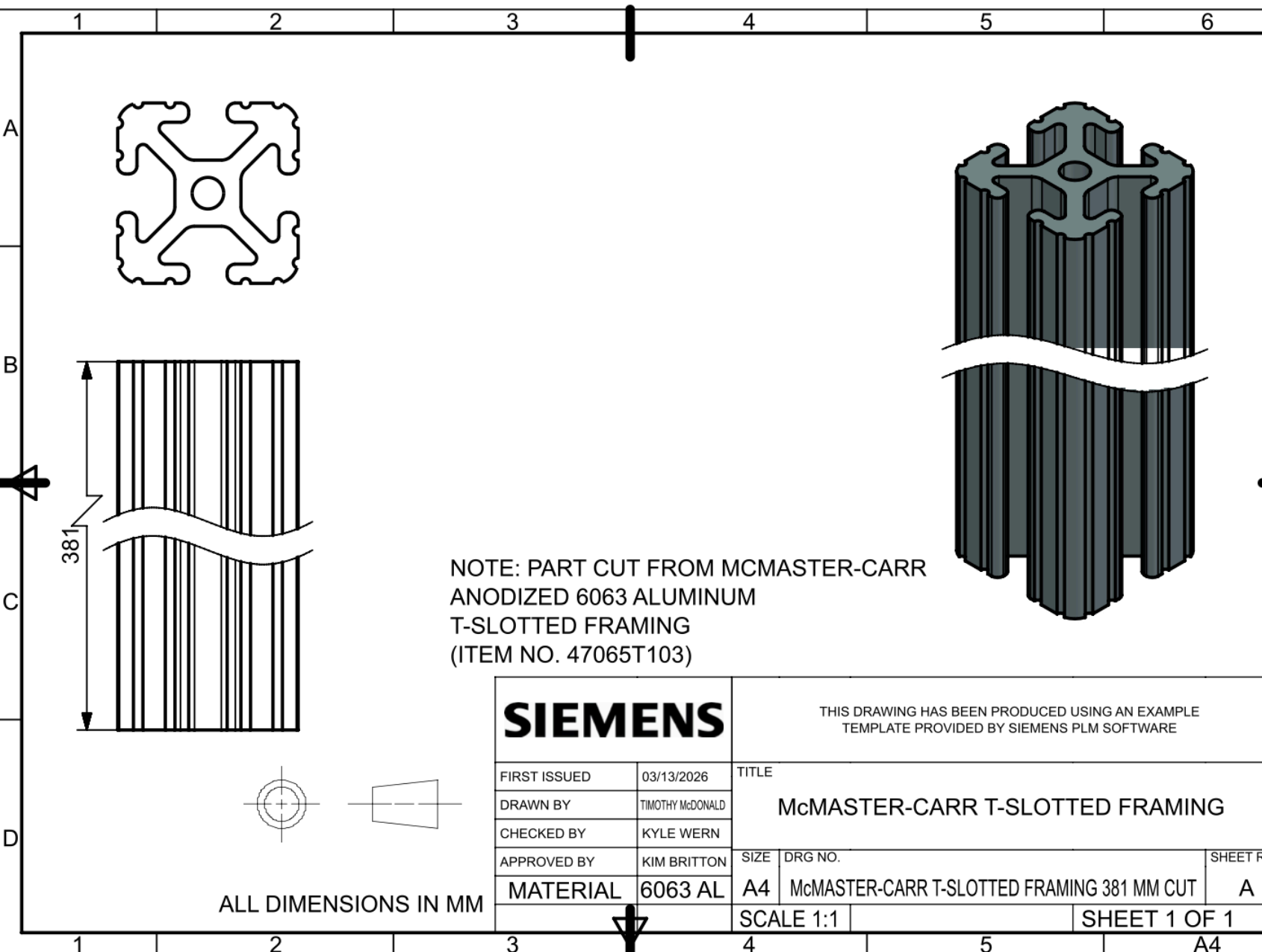
THIS DRAWING HAS BEEN PRODUCED USING AN EXAMPLE
TEMPLATE PROVIDED BY SIEMENS PLM SOFTWARE

FIRST ISSUED	03/13/2026
DRAWN BY	TIMOTHY McDONALD
CHECKED BY	KYLE WERN
APPROVED BY	KIM BRITTON
MATERIAL	6063 AL

TITLE			
McMASTER-CARR T-SLOTTED FRAMING			
SIZE	DRG NO.	SHEET REV	
A4	McMASTER-CARR T-SLOTTED FRAMING 1041 MM CUT	A	

ALL DIMENSIONS IN MM

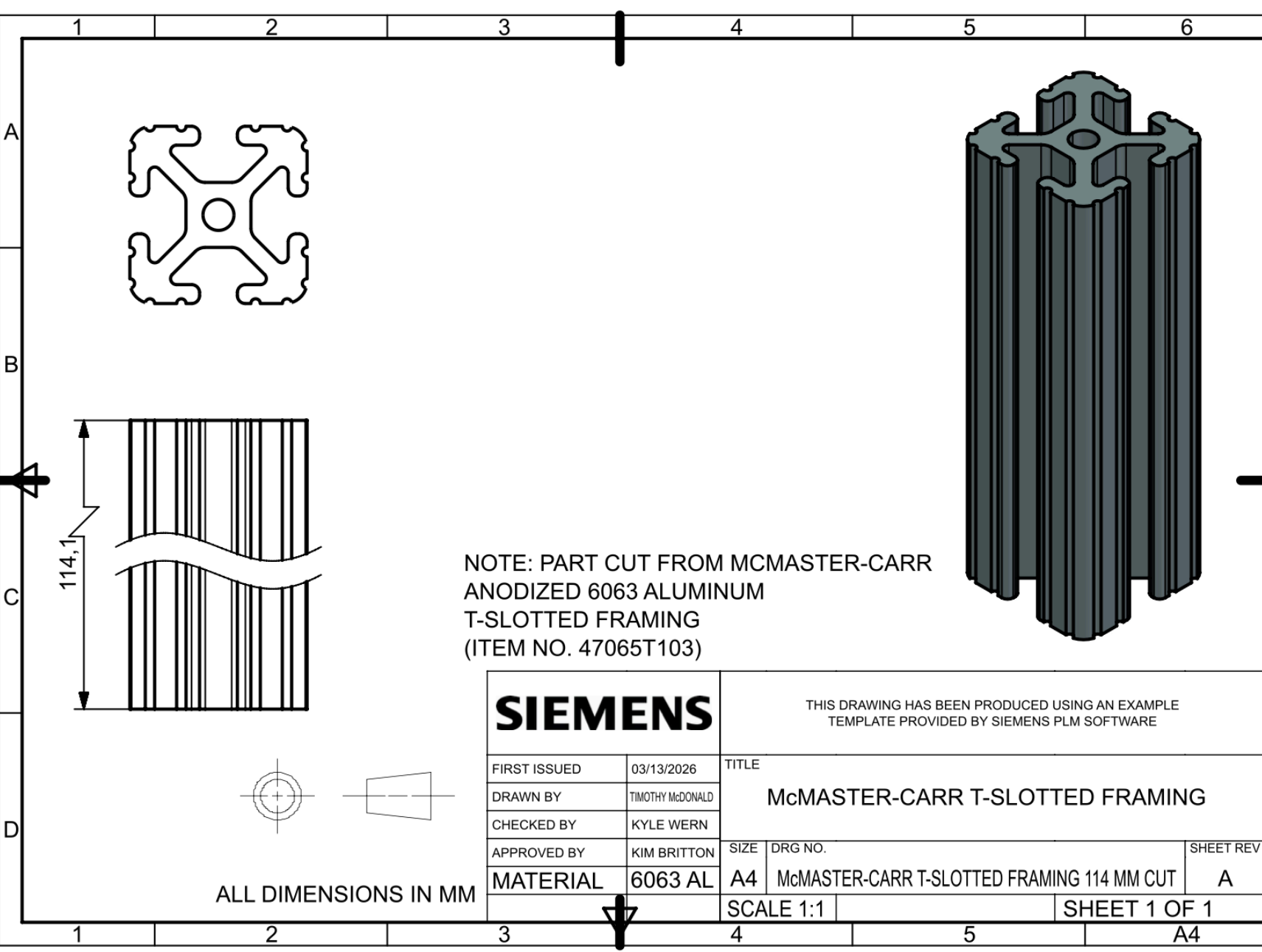
SCALE 1:1	SHEET 1 OF 1
-----------	--------------



NOTE: PART CUT FROM MCMaster-CARR
ANODIZED 6063 ALUMINUM
T-SLOTTED FRAMING
(ITEM NO. 47065T103)

SIEMENS		THIS DRAWING HAS BEEN PRODUCED USING AN EXAMPLE TEMPLATE PROVIDED BY SIEMENS PLM SOFTWARE			
FIRST ISSUED	03/13/2026	TITLE McMASTER-CARR T-SLOTTED FRAMING			
DRAWN BY	TIMOTHY McDONALD				
CHECKED BY	KYLE WERN				
APPROVED BY	KIM BRITTON	SIZE	DRG NO.	SHEET REV	
MATERIAL	6063 AL	A4	McMASTER-CARR T-SLOTTED FRAMING 381 MM CUT	A	
		SCALE 1:1		SHEET 1 OF 1	
				A4	

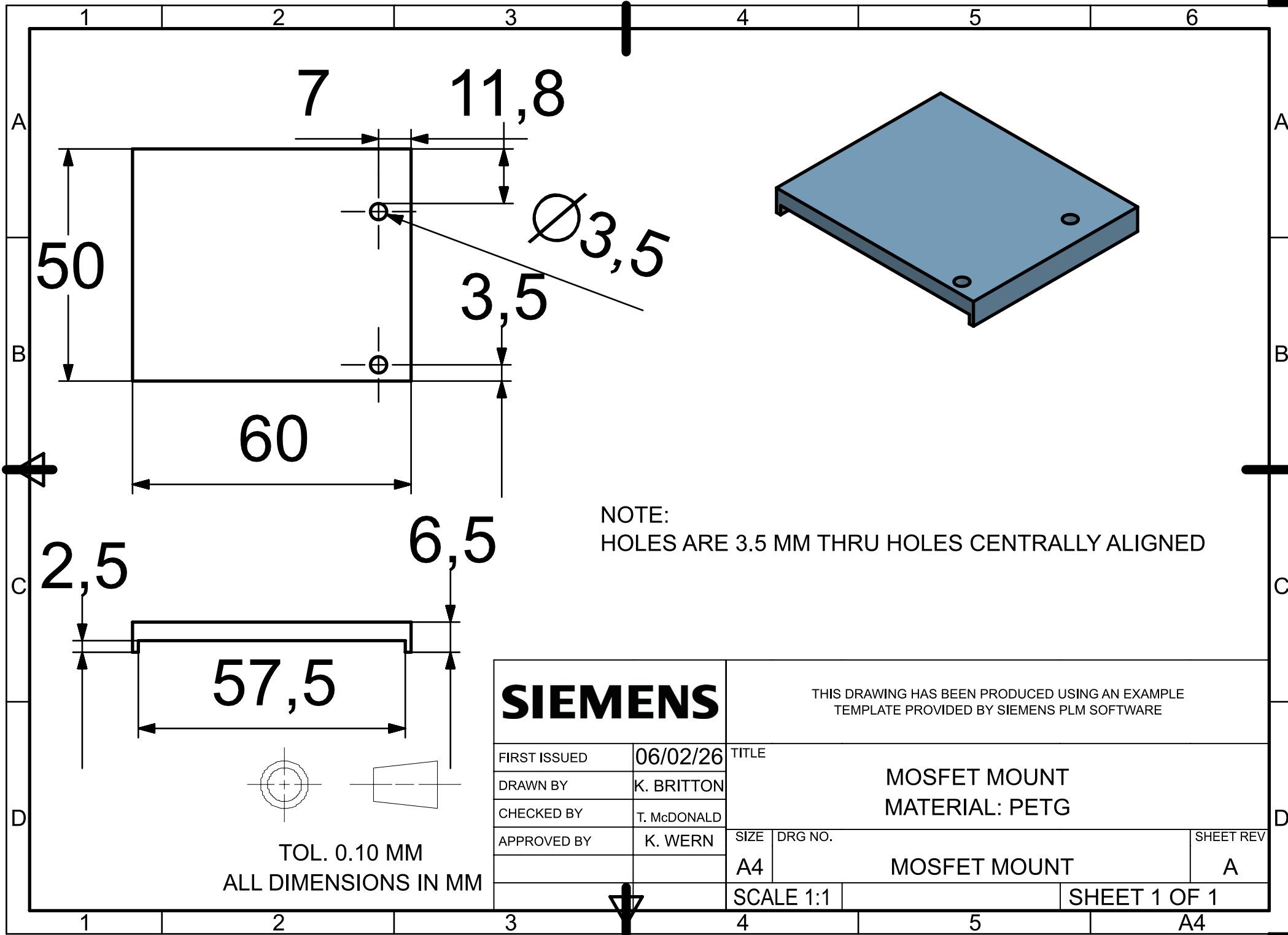
ALL DIMENSIONS IN MM

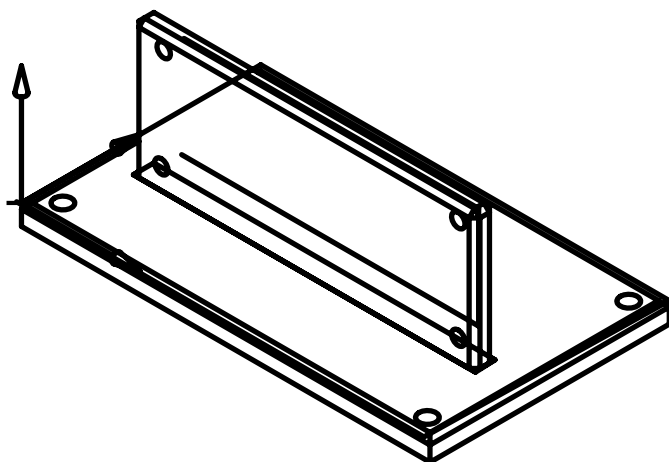
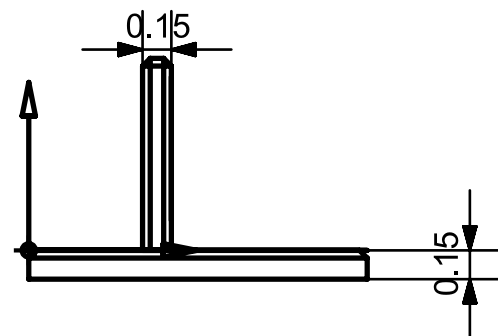
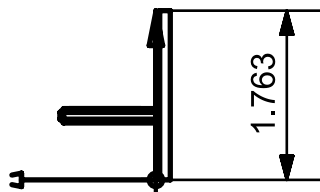
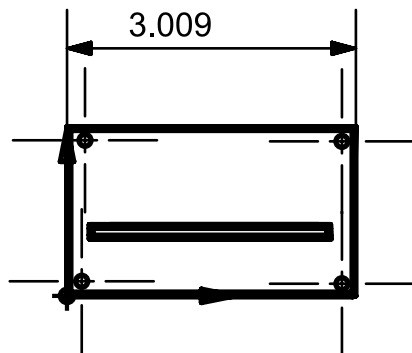


NOTE: PART CUT FROM MCMaster-CARR
ANODIZED 6063 ALUMINUM
T-SLOTTED FRAMING
(ITEM NO. 47065T103)

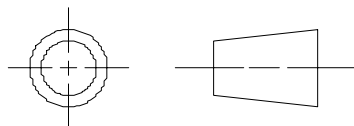
SIEMENS		THIS DRAWING HAS BEEN PRODUCED USING AN EXAMPLE TEMPLATE PROVIDED BY SIEMENS PLM SOFTWARE		
FIRST ISSUED	03/13/2026	TITLE McMASTER-CARR T-SLOTTED FRAMING		
DRAWN BY	TIMOTHY McDONALD			
CHECKED BY	KYLE WERN			
APPROVED BY	KIM BRITTON	SIZE	DRG NO.	SHEET REV
MATERIAL	6063 AL	A4	McMASTER-CARR T-SLOTTED FRAMING 114 MM CUT	A
		SCALE 1:1	SHEET 1 OF 1	

ALL DIMENSIONS IN MM





ALL DIMENSIONS IN INCHES
 MATERIAL: PETG
 TOLERANCE: +/-0.04 IN (1.0 MM) U.N.O.
 3D PRINTED COMPONENT



ALL DIMENSIONS IN INCHES

SIEMENS

THIS DRAWING HAS BEEN PRODUCED USING AN EXAMPLE
 TEMPLATE PROVIDED BY SIEMENS PLM SOFTWARE

FIRST ISSUED	06/03/2026
DRAWN BY	D. MARTIN
CHECKED BY	T. MCDONALD
APPROVED BY	K. WERN

TITLE		
PCA9685 MOUNT		
SIZE	DRG NO.	SHEET REV
A	capstonePCAmount1 (3)	A
SCALE 1:1		SHEET 1 OF 1

Artifact 6: Final Team Report - Design Changes

Required item: Final team report supporting document

Included file: team044-pianobot-design-changes.pdf

Design changes document included in the final team submission package.

Team 044 PianoBot - Summary of Design Changes

MIME Capstone Design II - MusicalBot Encore / PianoBot

Purpose

This document summarizes the major design changes represented by the final Team 044 PianoBot CAD package. It is included as iterative-project documentation for the current design relative to earlier MusicBot/GuitarBot-style concepts, low-fidelity prototypes, and intermediate CAD files.

Major design changes

- **Instrument platform:** The project moved toward a keyboard-based PianoBot that mounts over a Yamaha PSR-530 style 61-key instrument rather than relying on earlier autonomous instrument concepts.
- **Frame and structure:** Early proof-of-concept supports were replaced by a McMaster-Carr 80/20-style 6063 aluminum T-slotted frame. The drawing package includes cut-length drawings for the 1041 mm, 381 mm, and 114 mm frame pieces.
- **Actuation system:** Limited solenoid/key-group prototypes were expanded into a full solenoid bank using 25 N push-pull solenoids arranged into white-key and black-key octave assemblies.
- **Solenoid mounting plates:** Prototype mounting layouts were converted into custom 3D-printed mounting backplates for white-key 7-key, white-key 8-key, and black-key 5-key groups.
- **Electronics mounting:** The final package adds custom support documentation for the MOSFET mount and PCA9685 mount/support to organize control electronics and wiring.
- **Wiring and power distribution:** Terminal blocks, barrier blocks, wiring, power supplies, MOSFET modules, and PCA9685 boards are represented in the assembly drawings and/or the BOM.
- **Documentation consolidation:** The final submission consolidates previously separate Drive/Box/CAD exports into one drawing package PDF, a separate BOM spreadsheet, and this design-change summary.

Final deliverable note

The submitted drawing package contains assembly views, detailed subsystem/part drawings, and custom mount drawings. The separate BOM spreadsheet provides part names, source identifiers, descriptions, material/type notes, quantities, costs, and source links. Off-the-shelf parts are represented through the BOM and assembly views; custom or cut/manufactured parts are represented with detailed drawing sheets where available.

Artifact 7: Team Charter

Required item: Team charter

Included file: derek-martin-team-charter-clean.pdf

Cleaned team charter with revision markup removed.

Project Team Charter
MIME 498: Capstone 2
Signature Cover Page
ME- 044: Musicalbot Encore

I agree to do an equal amount of work in the team. I understand that my grade will reflect my effort in the team.

Printed Name: Kimberlee Britton

Signature: 

Written Statement:

I agree to do an equal amount of work in the team. I understand that my grade will reflect my effort in the team.

Printed Name: John Hansen

Signature: 

Written Statement:

I agree to do an equal amount of work in the team. I understand that my grade will reflect my effort in the team.

Printed Name: Tyler Kosa

Signature: 

Written Statement:

I agree to do an equal amount of work in the team. I understand that my grade will reflect my effort in the team.

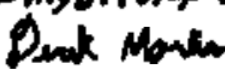
Printed Name: Cameron Lewis

Signature: 

Written Statement:

I agree to do an equal amount of work in the team. I understand that my grade will reflect my effort in the team.

Printed Name: Derek Martin

Signature: 

Written Statement:

I agree to do an equal amount of work in the team. I understand that my grade will reflect my effort in the team.

Printed Name: Tim McDonald

Signature: 

Written Statement:

I agree to do an equal amount of work in the team. I understand that my grade will reflect my effort in the team.

Printed Name: Kyle Wern

Signature: 

Written Statement:

I agree to do an equal amount of work in the team. I understand that my grade will reflect my effort in the team.

Team Purpose

As a team, we share a goal of bridging our passions for engineering, community outreach, and music. By creating a unique and highly interactive autonomously playing MusicalBot, we plan to engage K-12 students attending the 2026 Engineering Expo and inspire them to pursue a career in engineering. Through combining our skills in music creation, coding, and electronics, we aspire to create a functional prototype and memorable experience that students and their families will recall long after we graduate.

Our primary stakeholder for this project is Dr. Oman. Throughout this term, the course will challenge us to conceptualize and provide a final proof of concept we can manufacture in the last term of our undergraduate degrees. In addition to brainstorming, vetting, and finalizing a robust design for the MusicalBot, Dr. Oman expects us to document our project well and demonstrate the feasibility of our design. Additionally, to accomplish these goals, we recognize that we must communicate effectively and professionally with our stakeholders as well as with each other to meet deadlines and invent something truly remarkable.

Team Goals

Our primary goal is to finalize the design for a unique and interactive autonomous instrument that we can display at Expo. This device will not only produce music, but it will do so with an aesthetically pleasing appearance and diverse sound. Additionally, our plan to build the device from at least 50% recycled materials will reflect our commitment to sustainable and economically-conscious design. With these key factors in mind, our major goal will push us to demonstrate our resourcefulness and challenge us to think creatively with repurposed materials.

In addition to our main tasks, our “stretch goals” center around compatibility and open access. Rather than simply developing a machine that plays music alone, we strive to have it play alongside the existing GuitarBot and remain simplistic enough in design that students can adapt our findings to their own projects. These steps may include using an open-source coding language, adapting materials from basic electronics kits, and providing clear technical documentation that gives the information they need to implement our design process.

Because of our ambitious goals, our team is dedicated to upholding a high standard of performance and synchronization. We will use our time this term to strive for excellence in our design and earn As in the course. Along with collaborating on deliverables, we plan to remain accountable on our timecards and refer to this living document to recall our responsibilities.

Team Member Personalities and Roles

Our team's diverse High5 Strengths survey results provided us with insight into the strengths we bring to the table, which have been summarized in the list below.

- Kimberlee Britton: Catalyst, Empathizer, Timer Keeper, Deliverer, Analyst
- John (Jack) Hansen: Empathizer, Winner, Storyteller, Commander, Analyst
- Tyler Kosa: Believer, Problem Solver, Peace Keeper, Philomath, Coach
- Cameron Lewis: Thinker, Deliverer, Believer, Focus Expert, Catalyst
- Derek Martin: Focus Expert, Believer, Self-Believer, Philomath, Optimist
- Tim McDonald: Pacer Keeper, Philomath, Problem Solver, Empathizer, Analyst
- Kyle Wern: Problem Solver, Believer, Philomath, Analyst, Winner

Despite these data points, we felt at odds with some of our results and uncomfortable assigning roles without a deeper discussion. For this reason, we agreed to assign these roles by preference and be open to change. For now, our assumed positions and their responsibilities are summarized in the table below (see Table 1).

Role Title	Person(s)	Description
Project Manager	Kimberlee Britton Kyle Wern	Organize meetings with the team and the sponsor by providing agendas and taking notes Check that assignments and responsibilities are distributed equally across all team members
Logistics Manager	Jack Hansen	Make sure that all assignments are turned in on time and in accordance with the rubric
Financial Manager	Kyle Wern	Manage the finances and the ordering of materials as needed
Test Engineer	Cameron Lewis Jack Hansen Tim McDonald Tyler Kosa	Test the product at multiple stages
Manufacturing Engineer	Cameron Lewis Kyle Wern Kimberlee Britton	Create parts/ensure parts are designed to tolerance

	Tim McDonald Derek Martin Tyler Kosa Jack Hansen	
CAD/FEA Engineer	Tim McDonald Kyle Wern Tyler Kosa Cameron Lewis	Manage CAD modeling and dividing up the parts Manage Finite Element Analysis processing
Computer Engineer	Derek Martin Kimberlee Britton	Manage Python operations and MIDI simulations

Table 1: Team member assignments

Kimberlee Britton, one of our team's Project Managers, brings strong organizational and communication skills to the team by coordinating meetings with both the team and sponsor, preparing agendas, and documenting progress. With her attention to detail and ability to keep schedules on track, she ensures the entire team stays aligned with deadlines and project expectations, while also checking that assignments and responsibilities are distributed equally across all team members. She also serves as a Manufacturing Engineer, where she helps create parts and verify that designs are made to tolerance, and as a Computer Engineer, where she supports Python operations and MIDI simulations essential to our design process.

As a Computer Engineer, Derek Martin manages Python operations and MIDI simulations that support the team's design and testing process. His technical experience helps maintain the digital systems needed to evaluate concepts and support development efficiently. In addition, as a Manufacturing Engineer, he helps create parts and guarantees that designs are made to tolerance.

Tyler Kosa

serves as the test engineer, manufacturing engineer, and CAD engineer. His skills in analyzing mechanical components and CAD help the team create 3D printed parts and ensure our design's dimensions fall within tolerance.

With a strong mind for organization and deadlines, Jack Hansen, our Logistics Manager, verifies that assignments are submitted on time and in accordance with course rubrics. He monitors ongoing task completion, tracks required deliverables, and supports communication between members to maintain overall project flow.

Kyle Wern serves as the group's second Project Manager, Financial Manager, Manufacturing Engineer, and CAD/FEA Engineer. His enthusiasm for productive workflow; experience with budgeting, purchasing, and material sourcing; and interest in both manufacturing and CAD make him well-suited for these positions. His practical problem-solving skills and comfort with numbers allow him to handle cost decisions, evaluate material tradeoffs, and ensure that orders are placed efficiently. His skills in visualization, tolerance adherence, and design for manufacturing are also unmatched, making him an effective Manufacturing Engineer for our team.

As both the Test Engineer and a Manufacturing Engineer for the team, Cameron Lewis bridges the gap between design and production. He helps fabricate components, , assure that parts are designed to tolerance, and tests the product at multiple stages.

-He also serves as a CAD/FEA Engineer, where he helps manage CAD modeling, part division, and uses Finite Element Analysis to test parts' bending strength throughout the design process.

Tim McDonald serves as a Test Engineer, Manufacturing Engineer, and CAD/FEA Engineer for the team. In these roles, he tests the product at multiple stages, helps create parts and determines that they are designed to tolerance, and supports CAD modeling by helping divide up the parts across the team. He also contributes to Finite Element Analysis processing, helping strengthen the team's ability to evaluate and refine the design throughout development.

Ground Rules

To ensure we meet our deadlines on time, we have established regular meetings and communication channels. We hold meetings on Tuesdays from 4:00 – 6:00 pm and Fridays from

1:00 – 3:00 pm in a teammate’s garage. The team primarily uses the Tuesday meetings to work ahead on written deliverables due each Friday and Sunday, and to complete construction after each deliverable is at least 75% complete. Fridays’ meetings are similarly used for construction, as well as meeting with the team’s sponsor (Dr. Oman) and the rest of the Spring 2026 Capstone teaching team every other week for a 30-minute Zoom meeting. Throughout the week and over the weekend, each member of the team has also agreed to commit at least 2 hours outside of scheduled meeting hours toward construction and testing to make sure the project progresses at an optimal rate. Because of the hands-on nature of the meetings this term, the team prioritizes in-person meetings, and team members who cannot attend meetings in person due to travel or illness are expected to attend over Zoom or Discord to continue working with other team members on written deliverables.

Between our meetings, we communicate with one another primarily in our Discord channel to draft emails, ask clarifying questions, and send updates regarding our workflow. For more formal communication with our sponsor, we use Outlook and CC members on each email. In addition, we check both our emails and the Discord channel at least once a day to certify all parties (including our mentor/sponsor) are on the same page. As an example, for major design decisions or scope changes, prioritizing in-person check-ins with Dr. Oman before proceeding will ensure that our design remains aligned with sponsor expectations and reduces the risk of late-stage redesign.

In order to foster a healthy work environment, we have agreed to adhere to a few norms centered on mutual respect. In discussions, we treat everyone as we would like to be treated, acknowledge that every idea has potential, and agree that mistakes are a natural part of the engineering process. During discussions, we make sure each member can speak without interruption, and we listen actively to facilitate productive conversations. Along with respect, we also firmly believe in honesty. When dissenting views arise, we address them through in-person discussions where we focus on critiquing the idea rather than the person. By cultivating an environment where it is safe to disagree, we ensure that our final decisions are thoroughly evaluated and that every team member feels heard and respected.

To make sure the project stays on track, we apply several accountability measures centered on transparency and time management. During weekly meetings, we review progress and identify blockers early. If a task is taking longer than expected, it is reassigned or broken into smaller components. This approach allows for steady progress and prevents inefficient use of team resources. We utilize timecards weekly to track our efforts,

and spend the first 15 minutes of every team meeting reviewing each member's weekly progress. We expect a high level of commitment, meaning everyone attends all meetings unless a valid excuse arises, in which case they notify the group as early as possible. Most importantly, we rely on total honesty regarding timecard entries to maintain trust within the team and ensure that our reported data accurately reflects our collective labor.

Finally, we use a structured plan to manage the workload and maximize our efficiency. The Project Manager is responsible for identifying any tasks that cannot be completed on time and posting them for other members to pick up in our Discord channel thread "assignment links."

To make sure no one's time is wasted, we keep our meetings focused and productive by adhering to agendas and sticking to our project timelines. We also strive to distribute work in a way that aligns with each member's individual strengths and interests, making sure that everyone contributes in a way that is both effective for the project and personally engaging.

Potential Barriers and Coping Strategies

Although we are a cohesive team, we anticipate that some personal matters may affect the team's performance. As graduating seniors, our schedules are impacted by the day as we search for jobs and otherwise organize our post-graduation life. In order to reach our deadlines on time, we agree to communicate openly and honestly about our schedule limitations. For example, if a member cannot complete a task by a certain time due to personal events, that the member informs the team early so their work can be redistributed to other members. We also expect that this practice is used honestly and with the other team members in mind to avoid overwork. These expectations reflect some of our past project experiences, in which some members struggled with different teammates who did not use effective communication or follow through with their actions. In order to avoid these past cycles, we cultivate an environment where every member feels comfortable speaking openly, and we accomplish this goal by reinforcing the idea that every suggestion has value and must be met with respect. When we notice a drop in participation or a shift toward rude communication, we handle it through the in-person discussions outlined in our norms. These steps make certain that we act with integrity and address the behavior respectfully before it affects our collective progress.

Regarding our relationship with our sponsor, we anticipate that challenges like shifting schedules or differing perspectives on the project scope will arise. Since Dr. Oman has other academic and professional commitments, we remain flexible and diligent about checking in. To mitigate issues over misaligned timelines or the complexity of our group dynamics as we divide tasks, we communicate our schedules and project

planning with complete transparency. By reaching out as soon as a potential conflict or a technical disagreement occurs, we ensure that decisions are made collaboratively and that our progress remains unaffected by avoidable misunderstandings.

Prototyping Strategies

As we approach physical realization, we will test the feasibility of the design by relying on a balance of digital validation and team consensus. We plan to avoid scope creep by looking to our HoQ for reference and checking in with our sponsor. After creating our prototype in CAD, we will utilize FEA to simulate mechanical stresses and confirm that our design choices are sensible (e.g., that it can withstand multiple keys being pressed simultaneously). From here, we will meet as a team to review our final design, check with our sponsor that our design fits all criteria required, and order our materials. During this time,

we will continue using digital wiring diagrams to plan exactly what components we need to order, and how exactly each of these components will interact.

In order to avoid material procurement difficulties, we plan to order critical parts as early as possible and lean on reliable suppliers. By using sites like McMaster-Carr and Amazon Prime, we can minimize downtime. Throughout this stage, we will keep Dr. Oman updated via email or in class to ensure she remains informed of our technical progress and any issues we encounter obtaining materials. Additionally, we have scheduled bi-weekly meetings with Dr. Oman to ensure that we are on schedule and aligned with project goals.

If our prototype underperforms or fails, we will react by analysing its failures systematically. We intend to quantify any underperformance immediately, brainstorming solutions as a group to decide if the issue can be solved through iteration or if it indicates a fundamental design flaw. While we will "cross that bridge when we come to it" in terms of specific fixes, we are setting ourselves up now by ensuring all qualitative observations are recorded in our shared Google Drive. This documentation strategy will ensure that the knowledge we gain during assembly is preserved, allowing us to execute the most promising solutions quickly and keep the project moving forward. We will keep Dr. Oman informed about our prototyping process through the aforementioned bi-weekly meetings that have been scheduled, and will report to her about the current state of our prototyping efforts.

Our final prototyping strategy involves wiring a full white octave and black octave for testing with a sample MIDI file (e.g., the Pirates of the Caribbean theme). This wiring process will be completed by the end of the first week of the term. During this time, the team will verify

that each element required for the assembly has arrived or is in transit. This step will allow the team to complete Testing Procedures 1, 4, and 8, as outlined in our Design-based Testing Procedures document. From the results of these tests, the team can apply the lessons learned to lay the foundation for Testing Procedures 5, 6, and 7, which assess note accuracy, mechanical noise, and playable range. Once the team completes the aforementioned testing procedures for the two octave assemblies, we will move toward adapting what we have learned to create a fully integrated 61-key system. After creating the full prototype, the team will use Testing Procedures 2 and 9 to confirm playback reliability and that the device allows users to select multiple songs that play across the instrument fully. At this time, the team will ensure the exterior remains transparent to adhere to visibility requirements using Testing 3. When these testing procedures are complete, the team will complete Testing Procedures 10 and 11 to verify that the device can be set up in 10 minutes and transported by 2 team members. Lastly, the team will use Testing Procedures 12 and 13 to check that the user interface and poster are easy for a general audience to understand. With these procedures modeled after our HoQ in mind and regular check-ins with our sponsor, the team can move forward with creating an engaging and robust design capable of experiencing repeated use at the Spring 2026 Engineering Expo.

Artifact 8: Reflection Essay

Required item: Reflection essay mapping capstone experience to NACE competencies and future goals

Included file: derek-martin-reflection-essay-final.pdf

Final reflection essay PDF addressing all eight NACE competencies.

ePortfolio Reflection Essay Draft

Derek Martin | MIME 498 Capstone Design II | Oregon State University | May 2026

Introduction

The capstone project I am working on involves Musical Bot Encore, a modular robot designed to play household keyboard instruments. Compared to older exhibit concepts like GuitarBot, Musical Bot Encore is intended to be a more compact and portable device that can mount non-invasively on a household keyboard, perform timed mechanical key presses to play selected digital music tracks, and demonstrate mechanical, electrical, and software engineering in a way that K-12 students and families visiting Engineering Expo can see and hear. During capstone, I have been involved in technical writing through the design proposal, self-directed learning in Fusion 360, CAD modeling and drawing support for the frame and solenoid layout, and collaboration through team planning documents such as the team charter.

Reviewing the NACE career readiness competencies made me realize that my capstone project was much more than a technical task. My capstone was essentially a workplace simulation that required me to demonstrate effective communication, critical thinking, technology, teamwork, leadership, professionalism, career and self-development, and equity and inclusion. All of these competencies apply to my goal of pursuing a career in aerospace or mechanical engineering, where I will have to combine design, analysis, modeling, communication, and professional discipline.

Communication

Effective communication is especially noticeable in engineering projects because the project team must communicate a large amount of information to different audiences at various stages. In particular, the design proposal required me to explain the Musical Bot Encore project idea to both MIME faculty and the project sponsor. Additionally, the scope and constraints had to be defined clearly so the design would also make sense for the target audience of future Engineering Expo visitors.

The robot is not intended to be a full-fledged music-producing device that could be used for professional recording. Instead, Musical Bot Encore is intended to play pre-selected songs in a reliable way, mount non-invasively on a household instrument, have a simple and quick assembly procedure, and provide visitors with a clear connection between a solenoid moving and its corresponding sound.

As a result, capstone helped me become familiar with the importance of communication as a key engineering task. It is rare that calculations alone can guarantee the successful completion of a

mechanical or aerospace project because engineers constantly have to explain their assumptions and tradeoffs to people who may not have the same technical background. Preparing written material and developing the design with CAD models and drawings helped me prepare for communicating similar messages to engineers, managers, technicians, and potential clients in the future.

Critical Thinking

The application of critical thinking to engineering projects is best explained by the connection between requirements, constraints, and design evidence. A specific example from my own work is related to the actuation mechanism. We initially assumed that a MOSFET driver board was available, but the tests we conducted showed that the hardware in question was actually a relay board. The relay board could not handle the fast switching speeds required to play certain musical passages, so the team was forced to reconsider the design instead of proceeding with implementation based on a false premise.

The design proposal mentioned the need for MOSFET switching, PWM control, and a step-by-step approach to testing, moving from a single key to a solenoid bank and finally to the full assembly. Critical thinking proved to be important during capstone because it required me to examine assumptions carefully and make informed choices about feasibility. Similarly, my future engineering career will rely on the ability to distinguish between assumptions and evidence, especially when design decisions affect reliability, cost, safety, and performance.

Technology

Capstone helped me become familiar with the importance of technology by introducing self-learning and its application in a real project. Since my initial Fusion 360 skills were somewhat limited, I decided to spend about seven hours of self-directed learning getting comfortable with the basics of the program. I did not just watch the Product Design Online tutorials; I completed each of them and built practice parts including a Lego handle, glass bottle, paper clip, and complex bottle models.

Afterward, I used the same technology to create Fusion 360 assembly models and drawings that could be used to evaluate the design of the robotic piano player. Specifically, the mechanical model of the frame helped us consider the location of the solenoid bank, determine the size of the assembly, and make sure there was sufficient clearance for attaching the system to the keyboard.

This experience helped me understand that self-learning can be a very helpful engineering tool as long as I can establish a connection between new knowledge and a relevant design challenge. As a result, capstone helped me prepare for future engineering careers where constant learning will be essential.

Specifically, learning new CAD, simulation, controls, Python automation, and aerospace design software will be important as I move into mechanical and aerospace engineering work.

Teamwork

Capstone helped me better understand the necessity of teamwork because there is no point in designing, building, and demonstrating an electromechanical robot entirely by myself. In particular, teamwork was evident in our use of the team charter, which gave the project team a clear set of expectations, roles, communication procedures, working norms, and accountability measures.

The need for cooperation became evident because our tasks were highly interdependent. When one person adjusted the mechanical design, the rest of the team had to consider the impact of those changes on electronics, the mounting structure, solenoid bank position, and setup procedure. Creating a mechanical design model in CAD also made it easier for my teammates to discuss design decisions with each other because the model gave us a shared visual reference.

I believe that this experience with teamwork will prove valuable in my future career because engineering projects are multidisciplinary in nature. Aerospace and mechanical engineering often involve collaboration between designers, analysts, controls specialists, manufacturing specialists, quality staff, and operations staff. In addition, capstone helped me learn the importance of documenting the design process so individual work can support the whole team.

Leadership

Leadership during capstone was not tied only to titles. Instead, it referred to the necessity of realizing which actions were important and taking them to ensure that the project kept moving in the right direction. For instance, as the project started facing difficulties related to design evaluation and visualization, I took initiative by improving my Fusion 360 skills and using those skills to create mechanical design models for the frame and solenoid bank.

Another example of leadership can be associated with maintaining the right design focus and preventing scope creep. While it is tempting to try to build an autonomous robot capable of reproducing any type of music, our project has a simpler and more realistic goal: we are supposed to build a reliable, portable, and visually appealing exhibit that demonstrates engineering principles in a way that is understandable to K-12 students and families visiting Engineering Expo.

All things considered, I will be able to apply the knowledge and skills acquired in capstone to take initiative and demonstrate leadership within future engineering teams. My previous experience in

wildland firefighting and EMT training taught me the importance of responsibility, calm communication, and dependable performance under pressure. The capstone experience added an engineering context to those lessons by showing that leadership also requires technical judgment, initiative, and follow-through.

Professionalism

Professionalism is an important aspect of capstone because it refers to the necessity of building and using reliable engineering solutions for real problems. Since Musical Bot Encore is expected to be demonstrated publicly during Engineering Expo, we have to make sure the robot is reliable, easy to assemble, relatively light, portable, and non-invasive to a household keyboard. For instance, some of the constraints include a student budget of about \$1,000, setup time of no more than ten minutes, maximum weight of no more than 100 pounds, and the ability to mount non-invasively and be easily disassembled. These factors require us to think about the reliability and practicality of the solution rather than just its appeal.

Apart from constraints, I have gained an understanding of how professionalism manifests itself in the form of well-written and well-organized documents. The team charter established clear expectations for roles, communication, work norms, and accountability. The ePortfolio also required clear filenames, organized folders, and linked PDF or DOCX files so that reviewers could understand the work without needing extra explanation. I believe that these standards will prove valuable in my future engineering career, where I will have to pay close attention to details and ensure professionalism in documentation.

Career & Self-Development

Compared to some other competencies, I felt particularly comfortable with career and self-development during capstone because I was able to see gaps in my skills and address them. Capstone required me to learn new tools and apply them to a real project. Specifically, while I knew that CAD modeling would be required for mechanical and aerospace engineering roles, I did not have enough practical Fusion 360 skill before capstone. As a result, I devoted about seven hours to self-learning Fusion 360.

Once I learned the basics, I created mechanical design models and drawings for the robotic piano player. This example shows that I can use capstone as a way to acquire new skills on an ongoing basis and apply them to engineering challenges. In the future, I plan to continue graduate study in mechanical engineering at Oregon State University before working in mechanical or aerospace engineering positions that will allow me to demonstrate design, analysis, modeling, and communication skills.

Equity & Inclusion

Capstone introduced me to the idea of equity and inclusion both in relation to teamwork and outreach activities. Since I am working on Musical Bot Encore, I can consider the target audience of the project to be Engineering Expo visitors, including K-12 students and their families. Because of that audience, it made sense to make the robot visually attractive and understandable to people who may not know much about robotics or engineering.

For instance, the robot is not a hidden system with only a screen showing the process. Instead, the design is intended to make the physical action visible so visitors can observe the cause-and-effect relationship between mechanical movement and the sounds made by the keyboard. As far as teamwork is concerned, the team charter helped set ground rules and norms for working together, which supports inclusion by making expectations clear for people with different working styles.

While this approach helped me understand the importance of inclusion as a design criterion, I can still gain more insight into how engineering decisions include or exclude different users. As a mechanical or aerospace engineer, I will regularly face the challenge of designing for diverse user groups and promoting equitable access to engineering solutions. Capstone helped me see inclusion as a design principle rather than just a separate social topic.

Conclusion

Overall, capstone was an interesting opportunity that helped me practice multiple NACE competencies in a real-world engineering environment. In particular, the Musical Bot Encore project required me to communicate, think critically, learn new technologies, coordinate with teammates, assume leadership responsibility, remain professional, develop personally, and promote inclusion. These competencies apply to my future plans related to obtaining a master's degree and getting a job in aerospace or mechanical engineering.

Specifically, I will be able to use capstone as a platform for building the right skills and habits that will eventually make me ready for a professional engineering career. The most valuable thing I have realized during capstone is that career readiness can be achieved through a set of engineering practices. Specifically, I improved at learning, analyzing, documenting, collaborating, and communicating.

Source Reviewed

National Association of Colleges and Employers. Career Readiness Competencies for a Career-Ready Workforce. Revised 2024.